CYLINDRICALLY SYMMETRIC FRACTIONAL TRAVELING WAVES AND THEIR PARABOLOIDAL INTERFACES

Abstract. We construct cylindrically symmetric traveling waves for the Allen–Cahn equation with the full isotropic fractional Laplacian, for every order $0 < s < 1$, every positive speed and every ambient dimension at least two. The potential has two equal nondegenerate absolute minima; additional interior critical points and local wells are allowed. Every wave with the strict vertical and radial orders and the three pure-phase limits has an explicit leading interface asymptotic. In transverse dimension $n = 1$, the level height is equivalent to $cm_s r^{1+2s}/((1+2s)\kappa_s)$ at every fractional order. For $n \geq 2$, the height has order r^2 when $s > 1/2$, order $r^2/\log r$ when $s = 1/2$, and order r^{1+2s} when $s < 1/2$, with explicit coefficients independent of the level. Positive radially decreasing kernel approximations give the construction, and finite translation pairings handle all orders. A derivative-cone argument identifies the stationary layer in large-radius limits. Weighted layer matching and renormalized extension stress give an integrated radius balance. The force is determined by finite layer energy above $1/2$, a mesoscopic logarithm at $1/2$, and the fractional curvature of a transverse ball below $1/2$; in the plane it is the sharp axis energy at every order.

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1. Introduction

Equal well depths make a planar phase transition stationary, but they do not prevent a curved front from propagating. A cylindrical front can move vertically while its transition radius expands. The question is then whether the equation determines the growth of this radius and the shape of the level sets. We address this question for fractional diffusion. We determine the interface growth and its leading coefficient at every fractional order and in every ambient dimension. The distinction between finite layer energy, logarithmically divergent layer energy, and nonlocal curvature gives three different regimes when the transverse dimension is at least two. In the plane the same power $1 + 2s$ governs all orders.

We consider

$$u_t = -(-\Delta_{\mathbb{R}^{n+1}})^s u - f(u), \qquad (x, z) \in \mathbb{R}^n \times \mathbb{R}, \quad n \geq 1, \tag{1.1}$$

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with the potential assumptions

$$f = F' \in C^2(\mathbb{R}), \qquad F(-1) = F(1) = 0, \qquad F(t) > 0 \quad (t \neq -1, 1), \\ f'(-1) > 0, \qquad f'(1) > 0. \tag{A}$$

Only values near $[-1, 1]$ enter the arguments. There may be further critical points and local minima between the two absolute minima. The operator has Fourier symbol $|\zeta|^{2s}$, or $(|\xi|^2 + \eta^2)^s$ in the transverse and vertical dual variables. For $c > 0$, a traveling solution $u(x, z, t) = U(x, z - ct)$ has profile equation

$$L_{n+1,s}U - cU_z + f(U) = 0 \quad \text{in } \mathbb{R}^{n+1}, \qquad L_{d,s} = (-\Delta_{\mathbb{R}^d})^s. \tag{1.2}$$

1.1. **Main results.**

Theorem 1.1 (Cylindrical traveling waves)**.** *Assume* (A)*. For every $n \geq 1$, $0 < s < 1$ and $c > 0$, equation* (1.2) *has a bounded classical solution with $r = |x|$ such that*

$$-1 < U < 1, \qquad U(x, z) = U(r, z), \qquad U_z > 0, \qquad U_r < 0 \quad (r > 0), \tag{1.3}$$

and

$$\lim_{z \to -\infty} U(x, z) = -1, \qquad \lim_{z \to +\infty} U(x, z) = 1, \qquad \lim_{r \to \infty} U(r, z) = -1. \tag{1.4}$$

The vertical limits are locally uniform in x, and the lateral limit is locally uniform in z. For every $\theta \in (-1, 1)$ the level set $U = \theta$ is a C^2 radial graph $z = H_\theta(r)$, with $H'_\theta(r) > 0$ for $r > 0$ and $H_\theta(r) \to \infty$ as $r \to \infty$.

To state the geometric conclusions, let q be the increasing stationary layer

$$L_{1,s}q + f(q) = 0 \quad \text{in } \mathbb{R}, \qquad q(\pm\infty) = \pm 1.$$

It exists for all $0 < s < 1$ and is unique up to translation [2, Theorem 2.4]. For every $0 < s < 1$ put

$$m_s = \int_{\mathbb{R}} (q')^2 \, \mathrm{d}t, \qquad \kappa_s = \frac{2\Gamma(s + 1/2)}{\sqrt{\pi}\Gamma(1 - s)}. \tag{1.5}$$

For $n \geq 2$ and $0 < s < 1/2$, define also

$$\kappa_{n,s} = \frac{2\Gamma(n/2 + s)\Gamma(1/2 - s)}{\sqrt{\pi}\Gamma(1 - s)\Gamma(n/2 - s)}. \tag{1.6}$$

The mass m_s and the force coefficients are positive and finite in their stated ranges. When $1/2 < s < 1$, the layer also has the finite energy

$$\sigma_s = \frac{C_{1,s}}{4} \iint_{\mathbb{R}^2} \frac{(q(t) - q(t'))^2}{|t - t'|^{1+2s}} \, \mathrm{d}t \, \mathrm{d}t' + \int_{\mathbb{R}} F(q(t)) \, \mathrm{d}t, \qquad C_{1,s} = \frac{4^s \Gamma(1/2 + s)}{\sqrt{\pi}|\Gamma(-s)|}. \tag{1.7}$$

These constants are positive, finite and independent of the translation of q. The following results apply to every wave in the stated class.

Theorem 1.2 (Paraboloidal interfaces)**.** *Let $n \geq 2$, $1/2 < s < 1$, $c > 0$, and assume* (A)*. Every bounded classical solution of* (1.2) *satisfying* (1.3) *and the three phase limits* (1.4)*, with the local uniformity specified in Theorem 1.1, has*

$$\lim_{r \to \infty} \frac{H_\theta(r)}{r^2} = \frac{cm_s}{2(n - 1)\sigma_s} \qquad \text{for every fixed } \theta \in (-1, 1). \tag{1.8}$$

Equivalently, for the radius determined by $U(R_\theta(z), z) = \theta$ at large heights,

$$\lim_{z \to \infty} \frac{R_\theta(z)^2}{z} = \frac{2(n - 1)\sigma_s}{cm_s}. \tag{1.9}$$

Theorem 1.3 (Critical cylindrical interfaces)**.** *Let $n \geq 2$, $s = 1/2$, $c > 0$, and assume* (A)*. Every bounded classical solution of* (1.2) *satisfying* (1.3) *and* (1.4)*, with the local uniformity specified in Theorem 1.1, satisfies*

$$\lim_{r \to \infty} \frac{H_\theta(r) \log r}{r^2} = \frac{\pi c m_{1/2}}{4(n - 1)} \qquad \text{for every fixed } \theta \in (-1, 1). \tag{1.10}$$

Theorem 1.4 (Cylindrical interfaces below the half Laplacian)**.** *Let $n \geq 2$, $0 < s < 1/2$, $c > 0$, and assume (A). Every bounded classical solution of (1.2) satisfying (1.3) and (1.4), with the local uniformity specified in Theorem 1.1, satisfies*

$$\lim_{r\to\infty} \frac{H_\theta(r)}{r^{1+2s}} = \frac{cm_s}{(1+2s)\kappa_{n,s}} \qquad \text{for every fixed } \theta \in (-1,1). \tag{1.11}$$

Equivalently,

$$\lim_{z\to\infty} \frac{R_\theta(z)^{1+2s}}{z} = \frac{(1+2s)\kappa_{n,s}}{cm_s}.$$

Theorem 1.5 (Sharp planar interfaces)**.** *Let $n = 1$, $0 < s < 1$, $c > 0$, and assume (A). Every bounded classical solution of (1.2) satisfying (1.3) and (1.4), with the local uniformity specified in Theorem 1.1, satisfies*

$$\lim_{r\to\infty} \frac{H_\theta(r)}{r^{1+2s}} = \frac{cm_s}{(1+2s)\kappa_s} = \frac{cm_s\sqrt{\pi}\Gamma(1-s)}{2(1+2s)\Gamma(s+1/2)} \qquad (\theta \in (-1,1)). \tag{1.12}$$

In particular, at $s = 1/2$,

$$\lim_{r\to\infty} \frac{H_\theta(r)}{r^2} = \frac{\pi cm_{1/2}}{4}. \tag{1.13}$$

Equivalently, for every fixed $\theta \in (-1,1)$,

$$\lim_{z\to\infty} \frac{R_\theta(z)^{1+2s}}{z} = \frac{(1+2s)\kappa_s}{cm_s}.$$

The leading forces explain the dimension dependence. For $n \geq 2$ and $s > 1/2$, radial curvature contributes $(n-1)\sigma_s/R$. At $s = 1/2$, the layer energy accumulated up to the radius is logarithmic, giving $2(n-1)\log R/(\pi R)$. When $s < 1/2$, the whole transverse ball enters at scale R, and its fractional curvature gives $\kappa_{n,s}R^{-2s}$. The factor $n-1$ vanishes in the plane. The remaining interaction between the two sides of the front gives $\kappa_s R^{-2s}$, including at $s = 1/2$, where the planar law is quadratic without a logarithmic factor. The gamma expression in (1.6) reduces to κ_s when $n = 1$ and $s < 1/2$. All these laws concern each fixed level; no pointwise asymptotic formula for the slope of the radius is required.

1.2. **Relation to earlier traveling fronts.** For balanced local diffusion, Chen, Guo, Hamel, Ninomiya and Roquejoffre [5] constructed cylindrical traveling waves and obtained paraboloidal interfaces when $n \geq 2$; in the plane they obtained a hyperbolic-cosine growth law. Their result makes the role of the transverse geometry explicit. Fractional diffusion adds algebraic interaction with distant phases, so localization of the transition profile must be compatible with the full nonlocal equation.

One-dimensional fractional traveling waves and their speed identities were studied by Gui and Zhao [7], while the balanced stationary layer and its algebraic tails are provided by Cabré and Sire [1, 2]. In higher dimensions, Chan and Wei [4, Theorem 1.2] constructed fractional pyramidal fronts for $1/2 < s < 1$, starting from a planar front of positive speed k and taking $c > k$. Ma and Wang [8] established existence of cylindrically symmetric fractional fronts in $\mathbb{R}^3$. For the balanced potentials considered here, the one-dimensional speed identity forces an increasing transition between -1 and 1 to have zero speed. Our construction instead uses expanding half-cylinders and works for every positive prescribed c, every dimension $n+1 \geq 2$ and every $0 < s < 1$.

1.3. **The proof mechanisms.** The construction begins with a finite positive kernel obtained by capping the fractional kernel at the origin. Each transverse slice of this kernel is radially decreasing. Positive convolution and a scalar vertical transport formula therefore preserve radial decrease, vertical increase and radius order before the cap is removed. These orders provide local bounded variation compactness. At the lower end of a half-cylinder, sliding a stationary planar layer excludes every bounded stationary bubble with exterior value -1. At the upper end of the exhaustion, finite translation pairings exclude a nonconstant half-line limit. The latter argument uses the energy of an increment, which stays finite even when the total energy of a transition does not.

At every fractional order, a large-radius translation limit is planar, nonincreasing horizontally and nondecreasing vertically. A compact comparison profile gives a uniform lower bound for the horizontal derivative on the transition core. Comparing the vertical derivative with multiples of this positive derivative gives a least derivative cone. A positive least coefficient would produce an oblique monotone profile whose speed identity contradicts equal well depths. Thus the limit is the stationary layer. The same argument supplies qualitative profile convergence and horizontal phase and Cartesian derivative tails for the original wave in every dimension.

For the exact interface coefficient, first suppose $n \geq 2$ and $s > 1/2$, and let $R(z)$ be one fixed level radius. We integrate the radial extension stress on

$$r \geq h(z), \qquad h(z) = R(z)/2.$$

This cut has two simultaneous effects. Its boundary remains a distance comparable to $R(z)$ inside the upper phase, so its stress is $O(R^{-2s})$. On the retained region the curvature weight obeys $R(z)/r \leq 2$, so ordinary layer tightness identifies the curvature mass as $(\sigma_s + o(1))/R(z)$. Integration in height leaves the vertical coupling as a bounded endpoint flux. The integrated drift is $m_s R(z) + o(R(z))$, obtained from qualitative profile matching and monotonicity. Thus for $n \geq 2$ the balance becomes

$$cm_s R(z) + o(R(z)) = (n-1)\sigma_s \int_Z^z \frac{\mathrm{d}t}{R(t)} + o\left(\int_Z^z \frac{\mathrm{d}t}{R(t)}\right) + O(1).$$

Because $2s > 1$, the boundary error is smaller than the curvature term. The defining integral then gives (1.9); no derivative of a profile-matching error is needed. For the planar problem, the axis stress must instead be retained at its leading order. Rescaling by $R(z)$ gives the Poisson extension of $2\mathbf{1}_{(-1,1)} - 1$, whose axis energy is exactly κ_s. This identifies the coefficient of R^{-2s} for every $0 < s < 1$. At low orders the unweighted phase mass and the absolute stress flux may diverge. The primitive of $q'(q^{-1}(v))$ has integrable phase defects, and subtraction at one fixed height makes the flux convergent. Together they yield

$$cm_s R(z) + o(R(z)) = \kappa_s \int_Z^z R(t)^{-2s}\,\mathrm{d}t + o\left(\int_Z^z R(t)^{-2s}\,\mathrm{d}t\right),$$

which gives (1.12) by differentiating only the defining integral.

At $n \geq 2$ and $s \leq 1/2$, the same primitive gives the integrated drift outside $R/2$. The curvature integral remains finite with its radial weight, while subtracting the flux at one fixed height removes its large-extension-height divergence. For $s < 1/2$, rescaling both radius and extension height by R gives the extension of $2\mathbf{1}_{B_1 \subset \mathbb{R}^n} - 1$. Its finite energy first variation yields (1.6). At $s = 1/2$, the relevant scales are $1 \ll y \ll R$. The extension at each such scale approaches the half-space profile, whose squared derivative normal to the interface has integral $2/(\pi y)$. Integrating in y gives the logarithmic force and hence (1.10). In both cases the radius law follows from a cumulative forcing integral, without differentiating a profile-matching error.

1.4. **Outline of the paper.** Section 2 fixes the normalization and develops the translation and comparison tools. Section 3 constructs the ordered half-cylinder solutions and proves Theorem 1.1. Section 4 establishes phase decay, derivative-cone rigidity and the large-radius layer limit at every fractional order. Section 5 proves Theorem 1.2 by the moving-cut stress balance. Section 6 establishes the sharp axis-energy limit and the renormalized planar balance, proving Theorem 1.5. Section 7 uses the all-order preparation in arbitrary transverse dimension and proves Theorems 1.4 and 1.3 by the ball-force and mesoscopic limits. The interior and directional boundary estimates are proved in Appendix A; Appendix B constructs the positive massive resolvent used in the decay estimates.

2. Normalization and comparison principles

2.1. **The operator and the reference layer.** For $d \geq 1$ we fix the normalization

$$L_{d,s} v(X) = C_{d,s}\,\mathrm{PV} \int_{\mathbb{R}^d} \frac{v(X) - v(Y)}{|X - Y|^{d+2s}}\,\mathrm{d}Y, \qquad C_{d,s} = \frac{4^s \Gamma(d/2 + s)}{\pi^{d/2} |\Gamma(-s)|}. \tag{2.1}$$

Thus $\widehat{L_{d,s}v}(\xi) = |\xi|^{2s}\widehat{v}(\xi)$ for $\widehat{v}(\xi) = \int e^{-iX\cdot\xi}v(X)\,\mathrm{d}X$. For bounded locally C^2 functions the symmetric second-difference formula is absolutely convergent. In particular, coordinates on which the function does not depend may be integrated out: for $1 \le k \le d$ and $X \in \mathbb{R}^k \setminus \{0\}$,

$$C_{d,s}\int_{\mathbb{R}^{d-k}} (|X|^2 + |Y|^2)^{-(d+2s)/2}\,\mathrm{d}Y = C_{k,s}|X|^{-k-2s}. \tag{2.2}$$

The corresponding dimensional-reduction identity for $L_{d,s}$ holds distributionally by testing against compactly supported smooth functions. We write

$$K_{d,s}(v) = \frac{C_{d,s}}{4}\iint_{\mathbb{R}^d\times\mathbb{R}^d} \frac{(v(X)-v(Y))^2}{|X-Y|^{d+2s}}\,\mathrm{d}X\,\mathrm{d}Y$$

whenever this energy is finite. Its polarized operator form is

$$B_{d,s}(v,w) = \frac{C_{d,s}}{2}\iint \frac{(v(X)-v(Y))(w(X)-w(Y))}{|X-Y|^{d+2s}}\,\mathrm{d}X\,\mathrm{d}Y.$$

We use $H^s_0(D) = \{v \in H^s(\mathbb{R}^d) : v = 0 \text{ almost everywhere on } D^c\}$ for the zero-exterior energy space. A bounded classical solution means a bounded function on the full space, locally C^2 in its equation domain, for which the equation holds pointwise.

The minima in (A) satisfy $f(\pm1) = 0$. Choose once and for all $-1 < \alpha_- < \alpha < 1$ and $\lambda_0 > 0$ so that

$$f' \ge 2\lambda_0 \quad \text{on } [-1,\alpha_-] \cup [\alpha,1]. \tag{2.3}$$

The two intervals can be chosen to contain no interior root of f. Then $f(\alpha_-) > 0 > f(\alpha)$ and every root in $(-1,1)$ lies strictly between these levels. Constants below may depend on these fixed choices, n, s, c and the C^2 norm of f near $[-1,1]$.

The extension variable is denoted by y. Set $a = 1-2s$ and $d_s = 2^{2s-1}\Gamma(s)/\Gamma(1-s)$. For bounded v on $\mathbb{R}^d$ put

$$\begin{aligned} V(X,y) &= P^{(d)}_y * v(X), \\ P^{(d)}_y(X) &= \frac{\Gamma((d+2s)/2)}{\pi^{d/2}\Gamma(s)}\,\frac{y^{2s}}{(|X|^2+y^2)^{(d+2s)/2}}. \end{aligned} \tag{2.4}$$

The extension characterization of Caffarelli–Silvestre [3, Sections 2–3] gives

$$\operatorname{div}(y^a\nabla V) = 0, \qquad V(\cdot,0) = v, \qquad -d_s\lim_{y\downarrow0} y^a V_y = L_{d,s}v. \tag{2.5}$$

For bounded data the trace is understood at Lebesgue points and in distributions. The conormal equality is distributional for bounded data and locally uniform for bounded locally C^2 data. For finite homogeneous energy (without requiring $v \in L^2$),

$$K_{d,s}(v) = \frac{d_s}{2}\int_{\mathbb{R}^d}\int_0^\infty y^a|\nabla V|^2\,\mathrm{d}y\,\mathrm{d}X. \tag{2.6}$$

These constants will be kept throughout the stress calculations.

We will repeatedly use the following bounds. If v has bounded global C^2 norm, differentiation of (2.4) yields

$$|\nabla_X V| \le C\min(1,y^{-1}), \qquad |V_y| \le \begin{cases} Cy^{2s-1}, & 0 < y \le 1, \\ C/y, & y \ge 1. \end{cases} \tag{2.7}$$

The tangential bound follows by differentiating either the trace or the kernel, using $\|\nabla_X P^{(d)}_y\|_1 \le C/y$. Also $\|\partial_y P^{(d)}_y\|_1 \le C/y$ gives the normal bound for $y \ge 1$. One cancellation calculation proves both the remaining bound and the uniform conormal convergence needed on unbounded slices. Set $\beta = 2s$ and $N_y = -d_s y^{1-\beta}\partial_y P^{(d)}_y$. This kernel is even, has zero integral, and satisfies

$$|N_y(H)| \le C(|H|^2+y^2)^{-(d+\beta)/2}, \qquad N_y(H) \longrightarrow -C_{d,s}|H|^{-d-\beta} \quad (H \ne 0),$$

where the limiting constant follows from $2sd_s\Gamma((d+2s)/2)/(\pi^{d/2}\Gamma(s)) = C_{d,s}$. Consequently, writing $\Delta_H^2 v(X) = v(X+H)+v(X-H)-2v(X)$,

$$-d_s y^{1-\beta} V_y(X,y) = \frac{1}{2}\int_{\mathbb{R}^d} N_y(H)\Delta_H^2 v(X)\,\mathrm{d}H, \qquad |\Delta_H^2 v(X)| \le C\min\{|H|^2, 1\}.$$

The right-hand side is uniformly bounded: its absolute value is at most $C\int_0^1 r^{1-\beta}\,\mathrm{d}r + C\int_1^\infty r^{-1-\beta}\,\mathrm{d}r$. This proves $|V_y| \le Cy^{\beta-1}$ for $0 < y \le 1$. Moreover, the part with $|H| < \rho$ is at most $C\rho^{2-\beta}$, uniformly in X and y, for both the displayed integral and its limiting fractional-Laplacian integral. On $|H| \ge \rho$ the kernels converge in L^1 by dominated convergence. First letting $y \downarrow 0$ and then $\rho \downarrow 0$ proves

$$\sup_{X\in\mathbb{R}^d} \left|-d_s y^{1-2s} V_y(X,y) - L_{d,s}v(X)\right| \longrightarrow 0.$$

For a compact smooth test ψ, the bound $\|\Delta_H^2\psi\|_1 \le C_\psi \min\{|H|^2, 1\}$ gives $N_y * \psi \to L_{d,s}\psi$ in L^1 by the same argument; pairing with bounded v also justifies the distributional conormal limit in (2.5).

Lemma 2.1 (The reference layer)**.** *Under* (A)*, for every* $0 < s < 1$ *there is an increasing solution* q *of* $L_{1,s}q + f(q) = 0$*,* $q(\pm\infty) = \pm 1$*. It is unique up to translation, and* $q' > 0$*. We fix* $q(0) = \alpha$*. For all sufficiently large* $|t|$*,*

$$c_0|t|^{-2s} \le 1 - |q(t)| \le C|t|^{-2s}, \qquad 0 < q'(t) \le C|t|^{-1-2s}.$$

The mass m_s *in* (1.5) *is finite, strictly positive and invariant under translation for every* $0 < s < 1$*. If* $s > 1/2$*, the energy* σ_s *in* (1.7) *has the same properties. In this range, if* $Q = P_y^{(1)} * q$*, then*

$$F(q(t)) = \frac{d_s}{2}\int_0^\infty y^a(Q_t^2 - Q_y^2)\,\mathrm{d}y, \qquad \sigma_s = d_s\int_{\mathbb{R}}\int_0^\infty y^a Q_t^2\,\mathrm{d}y\,\mathrm{d}t. \tag{2.8}$$

Proof. Apply Cabré–Sire [2, Theorems 2.4 and 2.7] with their reaction equal to $-f$ and their potential equal to F. The required strict inequality between the potential and its endpoint values is exactly (A); also $F'(\pm 1) = 0$ and $(-f)'(\pm 1) < 0$. Their $C^{1,\gamma}$ condition is met by choosing $\max(0, 1-2s) < \gamma < 1$ and multiplying the reaction by a smooth compactly supported cutoff equal to one near $[-1, 1]$ if necessary. The solution takes values in $(-1, 1)$, so this modification changes neither the equation nor the layer. Interior local wells are allowed in their Remark 2.6.

The derivative bounds imply $q' \in L^1 \cap L^\infty \cap L^2$. For a translation by h, the squared L^2 difference is at most $h^2\|q'\|_2^2$ when $|h| \le 1$, and at most $4|h|$ for $|h| \ge 1$, since monotonicity gives $\|q(\cdot+h) - q\|_1 = 2|h|$ and $\|q(\cdot+h) - q\|_\infty \le 2$. Integration against $|h|^{-1-2s}$ gives the convergent integrals $\int_0^1 h^{1-2s}\,\mathrm{d}h$ and $\int_1^\infty h^{-2s}\,\mathrm{d}h$ when $2s > 1$. Taylor expansion at the two nondegenerate wells gives $F(q(t)) \le C(1+|t|)^{-4s}$. These estimates prove finiteness; nonconstancy gives positivity. Translation invariance follows by a change of variables. Apply the one-dimensional Hamiltonian identity [1, Theorem 2.3(i), Remark 3.11(b)] in the normalization (2.5). In their $(1+a)$ conormal convention take reaction $\widetilde{f} = -(1+a)f/d_s$ and potential $\widetilde{F} = (1+a)F/d_s$; cancellation of the common factor gives the first equality in (2.8). Adding its integral to (2.6) gives the second. □

2.2. **Finite translation tests.** For functions on the line, write $D_h v(t) = v(t+h) - v(t)$ and $K_s = K_{1,s}$, $B_s = B_{1,s}$. Equivalently,

$$\begin{aligned} K_s(v) &= \frac{C_{1,s}}{4}\int_{\mathbb{R}} \frac{\|D_r v\|_2^2}{|r|^{1+2s}}\,\mathrm{d}r, \\ B_s(v,\psi) &= \frac{C_{1,s}}{2}\int_{\mathbb{R}} \frac{\int_{\mathbb{R}} D_r v(t) D_r\psi(t)\,\mathrm{d}t}{|r|^{1+2s}}\,\mathrm{d}r. \end{aligned} \tag{2.9}$$

The mixed form can be finite even when $K_s(v)$ is infinite.

Lemma 2.2 (Finite-increment pairing)**.** *Let* $0 < s < 1$ *and let* v *be bounded, continuous and monotone on* $\mathbb{R}$*. Set*

$$A = |v(+\infty) - v(-\infty)|, \qquad \omega(r) = \sup_{|t-t'|\le r} |v(t) - v(t')|,$$

and assume

$$\int_0^1 \omega(r) r^{-2s}\,\mathrm{d}r < \infty. \tag{2.10}$$

For every h, $D_h v \in H^s(\mathbb{R}) \cap L^1(\mathbb{R})$ and

$$\|D_h v\|_1 = A|h|, \qquad \|D_h v\|_\infty \le \omega(|h|), \qquad \|D_h v\|_2^2 \le A|h|\omega(|h|). \tag{2.11}$$

For this fixed v, $B_s(v,\cdot)$ is absolutely convergent and continuous on $H^s \cap L^1$ with norm $\|\cdot\|_{H^s}+\|\cdot\|_1$, and agrees with the distribution $L_{1,s}v$ on compact smooth tests. Moreover, $D_h v$ is continuous and vanishes at both infinities, and there are $\psi_j \in C_c^\infty(\mathbb{R})$ such that

$$\|\psi_j - D_h v\|_{H^s} + \|\psi_j - D_h v\|_1 + \|\psi_j - D_h v\|_\infty \longrightarrow 0.$$

If $D_h v = 0$ on $[a,\infty)$, these approximations may be chosen in $C_c^\infty((-\infty,a))$. Finally,

$$B_s(v, D_h v) = -K_s(D_h v). \tag{2.12}$$

Proof. The derivative measure of v has total variation A. Integrating its mass on intervals of length $|h|$ by Tonelli gives the first identity in (2.11); the other two follow from the definition of the modulus. In particular,

$$\int_{|r|<1} \frac{\|D_r v\|_2^2}{|r|^{1+2s}}\,\mathrm{d}r \le 2A \int_0^1 \omega(r) r^{-2s}\,\mathrm{d}r < \infty. \tag{2.13}$$

Write $\delta_h = D_h v$. Since $D_r\delta_h = (D_r v)(\,\cdot + h) - D_r v$, its small-displacement energy is at most four times (2.13). For $|r| \ge 1$, use $\|D_r\delta_h\|_2 \le 2\|\delta_h\|_2$. Thus $\delta_h \in H^s$.

Let E_v denote the finite integral on the left of (2.13). Cauchy–Schwarz in (r,t) bounds the part of the absolute pairing integral with $|r| < 1$ by $C_s E_v^{1/2}\|\psi\|_{H^s}$. For the remaining part,

$$\int_{\mathbb{R}} |D_r v D_r \psi|\,\mathrm{d}t \le 4\|v\|_\infty \|\psi\|_1.$$

Consequently the pairing is absolutely convergent and

$$|B_s(v,\psi)| \le C_s E_v^{1/2}\|\psi\|_{H^s} + C_s\|v\|_\infty\|\psi\|_1,$$

which proves continuity in the asserted norm. For $\psi \in C_c^\infty(\mathbb{R})$, symmetric differences give

$$\|L_{1,s}\psi\|_1 \le \frac{C_{1,s}}{2}\int_{\mathbb{R}} \frac{\min\{|r|^2\|\psi''\|_1, 4\|\psi\|_1\}}{|r|^{1+2s}}\,\mathrm{d}r < \infty, \qquad \int_{\mathbb{R}} L_{1,s}\psi\,\mathrm{d}t = 0.$$

The zero integral follows by Fubini, since each translated second difference has zero integral. Omitting $|r| < \varepsilon$ and symmetrizing identifies the truncated pairing with $\int v L_{1,s}^{\varepsilon}\psi$. The same second-difference bound gives $L_{1,s}^{\varepsilon}\psi \to L_{1,s}\psi$ in L^1; absolute convergence of the pairing therefore proves $B_s(v,\psi) = \int v L_{1,s}\psi$.

For each fixed r, the function $D_r v$ belongs to L^2. Translation invariance of its squared norm gives

$$2\langle D_r v, D_r\delta_h\rangle + \|D_r\delta_h\|_2^2 = 0.$$

Integration with weight $C_{1,s}/(4|r|^{1+2s})$ proves (2.12). Every quantity in this step is finite; no difference of the energies of v and its translate is taken.

It remains to justify the simultaneous approximation. The finite endpoint limits of v imply $\delta_h \in C_0(\mathbb{R})$, the space of continuous functions vanishing at infinity. For any $g \in H^s \cap L^1 \cap C_0$, take $\chi_R(t) = \chi(t/R)$, where $0 \le \chi \le 1$ is smooth, compactly supported and equal to one on $[-1,1]$. Then $\chi_R g \to g$ in L^1, L^2 and uniformly. To check the Sobolev seminorm, write

$$D_r[(1-\chi_R)g](t) = (1-\chi_R(t+r))D_r g(t) - g(t)D_r\chi_R(t).$$

The weighted square integral of the first term tends to zero by dominated convergence, since $g \in H^s$. For the second it is at most

$$C\|g\|_2^2 \int_{\mathbb{R}} \frac{\min\{1, |r|^2/R^2\}}{|r|^{1+2s}}\,\mathrm{d}r \le CR^{-2s}\|g\|_2^2 \longrightarrow 0.$$

Mollification then converges in all three asserted norms, by translation continuity in H^s and L^1 and uniform continuity of C_0 functions. If $g = 0$ on $[a, \infty)$, first replace it by $g(\cdot + \varepsilon)$; its support lies in $(-\infty, a - \varepsilon]$. Apply the cutoffs and mollify at radius less than $\varepsilon/2$, then choose a diagonal sequence as $\varepsilon \downarrow 0$. This proves the support-preserving assertion. □

Lemma 2.3 (A modulus from the boundary powers)**.** *Suppose that v is bounded and continuous on $\mathbb{R}$, vanishes on $[0, \infty)$, and is C^1 on $(-\infty, 0)$. If, for some $0 < p < 1$,*

$$|v(-r)| \le Cr^p, \qquad |v'(-r)| \le Cr^{p-1} \quad (0 < r < 1), \qquad \sup_{t<-1/2} |v'(t)| \le C,$$

then $v \in C^{0,p}(\mathbb{R})$. Its modulus satisfies (2.10) whenever $p > 2s - 1$. For $s < 1/2$, boundedness alone implies this integral condition.

Proof. Boundedness handles distances $d \ge 1/4$. For two negative points at distances $r, r + d$ from zero, a bounded derivative gives Cd^p when $r \ge 1/2$. If $r < 1/2$, then $r + d < 3/4$: for $r \le 2d$ use the two value bounds, giving $C((2d)^p + (3d)^p)$; for $r > 2d$ integrate the derivative to obtain $Cdr^{p-1} \le Cd^p$. Crossing pairs use the power bound and zero exterior value; two nonnegative points have equal values. Thus $\omega(d) \le Cd^p$, and its weighted integral converges when $p > 2s - 1$. For $s < 1/2$, the bound $\omega \le 2\|v\|_\infty$ suffices. □

Lemma 2.4 (Exclusion on the half-line)**.** *Let $0 < s < 1$, $c > 0$, and $f = F' \in C^1(\mathbb{R})$, with $F(1) = 0$ and $F \ge 0$ on $[-1, 1]$. Suppose that $w : \mathbb{R} \to [-1, 1]$ is continuous, nondecreasing, C^1 on $(-\infty, 0)$, equal to 1 on $[0, \infty)$, and satisfies*

$$L_{1,s}w - cw' + f(w) = 0 \quad \text{in } \mathcal{D}'((-\infty, 0)).$$

If its modulus satisfies (2.10), then $w \equiv 1$.

Proof. Let $v = 1 - w$, $\beta = w(-\infty)$, and $A = 1 - \beta$. Monotonicity and the fundamental theorem of calculus on compact negative intervals, followed by the continuous trace at zero, give $v' \in L^1(\mathbb{R})$, $v' \le 0$, and $\|v'\|_1 = A$. The distributional derivative has no atom at zero. The equation becomes

$$L_{1,s}v - cv' = f(1 - v) \quad \text{on } (-\infty, 0).$$

For $h > 0$, $\delta_h = D_h v \le 0$ vanishes on $[0, \infty)$. Lemma 2.2 supplies compact smooth tests in $(-\infty, 0)$ converging to δ_h in H^s, in L^1, and uniformly. The fractional term passes by continuity of the pairing, the drift by uniform convergence against $v' \in L^1$, and the reaction by L^1 convergence against bounded $f(1 - v)$. Applying (2.12) and dividing by $h > 0$ gives

$$0 \ge -\frac{K_s(\delta_h)}{h} = c \int_{\mathbb{R}} v' \frac{\delta_h}{h} \, \mathrm{d}t + \int_{\mathbb{R}} f(1 - v) \frac{\delta_h}{h} \, \mathrm{d}t. \tag{2.14}$$

The difference quotient is the average $\delta_h/h = \int_0^1 v'(\cdot + th) \, \mathrm{d}t$. It converges to v' in L^1 and pointwise in the open half-line. The reaction term therefore tends to

$$\int_{\mathbb{R}} f(1 - v)v' \, \mathrm{d}t = -\big[F(1 - v)\big]_{-\infty}^{+\infty} = F(\beta),$$

by the absolutely continuous chain rule. The first integrand in (2.14) is nonnegative. Fatou's lemma yields

$$0 \ge c \int_{\mathbb{R}} |v'|^2 \, \mathrm{d}t + F(\beta) \ge 0.$$

Consequently $v' = 0$ almost everywhere, and its zero trace at the origin gives $v \equiv 0$. □

For the half-line limits arising from the cylinder construction, the modulus condition is available at every order. If $s < 1/2$, use boundedness. At $s = 1/2$, the top power estimate of Lemma A.2 with any $0 < p < 1/2$, followed by Lemma A.3, gives the required modulus. If $s > 1/2$, choose

$$2s - 1 < p < s.$$

The interval is nonempty because $s < 1$. Lemma A.2 gives the value and derivative powers for $v = 1 - w$, with bounded forcing $f(1 - v)$; interior regularity gives the derivative bound away

from zero. Lemma 2.3 then proves (2.10). For the critical application $f \in C^2$ and $f(1) = 0$ are supplied by (A).

Lemma 2.5 (The whole-line speed identity)**.** *Let $0 < s < 1$, $c > 0$, and $f = F' \in C^1(\mathbb{R})$. If $\varphi : \mathbb{R} \to [-1, 1]$ is nondecreasing, C^1 with bounded derivative, and*

$$L_{1,s}\varphi - c\varphi' + f(\varphi) = 0 \quad \text{in distributions,}$$

then its endpoint limits $\beta_\pm$ satisfy

$$f(\beta_\pm) = 0, \qquad c\int_{\mathbb{R}} (\varphi')^2 \, \mathrm{d}t = F(\beta_+) - F(\beta_-). \tag{2.15}$$

Proof. Monotonicity gives $\int \varphi' = \beta_+ - \beta_-$; together with the bounded derivative this proves $\varphi' \in L^1 \cap L^2$. The Lipschitz modulus satisfies (2.10) because $s < 1$. Choose $\psi \in C_c^\infty(\mathbb{R})$ with $\int \psi = 1$. The compact-test calculation in Lemma 2.2 gives $L_{1,s}\psi \in L^1$ and $\int L_{1,s}\psi = 0$. For $\varphi_a(t) = \varphi(t + a)$, the translated weak equation reads

$$\int_{\mathbb{R}} \varphi_a L_{1,s}\psi \, \mathrm{d}t + c\int_{\mathbb{R}} \varphi_a \psi' \, \mathrm{d}t + \int_{\mathbb{R}} f(\varphi_a)\psi \, \mathrm{d}t = 0.$$

As $a \to \pm\infty$, bounded convergence applies to all three terms. The first two tend to zero because $\int L_{1,s}\psi = \int \psi' = 0$, and the third tends to $f(\beta_\pm)$, proving the endpoint assertion.

For $h \neq 0$, Lemma 2.2 provides $\delta_h = D_h\varphi \in H^s \cap L^1$ and its simultaneous approximation by compact smooth tests. The fractional term passes by continuity of the pairing, the drift by L^2 convergence against $\varphi' \in L^2$, and the reaction by L^1 convergence against bounded $f(\varphi)$. Thus (2.12) yields

$$\int_{\mathbb{R}} f(\varphi)\frac{\delta_h}{h} \, \mathrm{d}t = c\int_{\mathbb{R}} \varphi' \frac{\delta_h}{h} \, \mathrm{d}t + \frac{K_s(\delta_h)}{h}.$$

The identity $\delta_h/h = \int_0^1 \varphi'(\cdot + \tau h) \, \mathrm{d}\tau$ and continuity of translations show that these quotients converge in $L^1 \cap L^2$ to φ' as $h \to 0$ from either side. The absolutely continuous chain rule identifies the reaction limit. The first two terms therefore tend to $F(\beta_+) - F(\beta_-)$ and $c\|\varphi'\|_2^2$. The last term is nonnegative for $h > 0$ and nonpositive for $h < 0$; the opposite one-sided inequalities give (2.15). □

2.3. **Compact profiles and stationary comparison.**

Lemma 2.6 (Compact comparison profiles)**.** *Assume (A). For every $d \geq 1$, $0 < s < 1$, and $\gamma < 1$, there exist $L > 0$, a C^2 reaction g on $[-1, 1]$, and a radial nonincreasing function $b : \mathbb{R}^d \to [-1, 1]$ with $b \in C(\mathbb{R}^d) \cap C^2(B_L)$ such that*

$$b = -1 \text{ on } B_L^c, \qquad -1 < b < 1 \text{ on } B_L, \qquad b(0) > \gamma, \qquad L_{d,s}b + g(b) = 0 \text{ in } B_L. \tag{2.16}$$

The reaction can be chosen arbitrarily close to f in C^1, with $g(\pm 1) = 0$ and $g > f$ on $(\alpha_-, 1)$. If $c \geq 0$ and v is a bounded classical solution of $L_{d,s}v - cv_z + f(v) = 0$, where z is a distinguished coordinate, with $v \geq b$ globally and $v_z \geq 0$, then an interior contact with common value above α_- is impossible.

Proof. We first favor the well at 1 without changing the endpoint reactions. Choose $h \in C^\infty([-1, 1])$ positive inside and vanishing to second order at the endpoints, and choose $\zeta \in C_c^\infty((-1, \alpha_-))$ nonnegative with $\int \zeta > \int h$. For sufficiently small fixed $\delta > 0$, set

$$g = f + \delta(h - \zeta), \qquad G(t) = F(t) + \delta\int_1^t (h - \zeta)(\tau) \, \mathrm{d}\tau.$$

Then $G(1) = 0 < G(-1)$ and 1 is the unique minimizer of G on $[-1, 1]$. To verify positivity, put $J(t) = \int_1^t (h - \zeta)$ and $A = J(-1) = \int_{-1}^1 (\zeta - h) > 0$. Near -1, $\zeta = 0$ and $J(t) \geq A$. Near 1, $|J(t)| \leq C(1 - t)^3$, whereas $F(t) \geq c(1 - t)^2$. On the remaining compact interior interval, F has a positive minimum. These three estimates give $G(t) > 0$ for every $-1 \leq t < 1$ once δ is small enough. Also $g - f = \delta h > 0$ on $(\alpha_-, 1)$ and $g'(\pm 1) = f'(\pm 1) > 0$. The constructed reaction is C^2, and $\|g - f\|_{C^1} \to 0$ as $\delta \downarrow 0$. Fix such a δ for the rest of the construction.

Extend G to a coercive C^3 function on the real line, quadratic at infinity, with bounded second derivative and increasing away from each endpoint outside $[-1,1]$. Such an extension is obtained by continuing the positive second derivatives near the endpoints and then interpolating to a positive constant. Clipping to $[-1,1]$ decreases this potential. Minimize

$$K_{d,s}(b)+\int_{B_L}[G(b)-G(-1)]\,\mathrm{d}X, \qquad b+1\in H^s_0(B_L).$$

After adding $G(-1)|B_L|$, both terms are nonnegative. Clipping decreases the kinetic term as well as the potential. For a clipped minimizing sequence, $0\le b+1\le 2$ with support in $\overline{B}_L$, so its L^2 norm is bounded. The kinetic bound therefore gives an H^s bound. Weak compactness and compact embedding on the fixed ball give a limit in $H^s_0(B_L)$, strong L^2 convergence and, after a subsequence, almost-everywhere convergence. The kinetic term is weakly lower semicontinuous and the bounded-range potential integral converges by dominated convergence. This proves existence of a minimizer with values in $[-1,1]$.

Schwarz rearrangement of $b+1$ preserves its potential and does not increase the kinetic term. To see the latter directly, replace the singular kernel by the integrable radially decreasing cap $\min\{\varepsilon^{-d-2s},|X|^{-d-2s}\}$. Expand the square for $b+1\in L^2$, whose differences equal those of b. The two square terms depend only on its L^2 norm, and the mixed term increases by the Riesz rearrangement inequality. Monotone convergence removes the cap. We may consequently take the minimizer radial and nonincreasing. Because the potential was extended before minimization, the clipped minimizer still minimizes an unconstrained functional. Arbitrary signed compact variations give

$$B_{d,s}(b,\varphi)+\int_{B_L} g(b)\varphi\,\mathrm{d}X=0, \qquad \varphi\in C^\infty_c(B_L),$$

with no obstacle multiplier.

Apply bounded-forcing fractional Dirichlet regularity [9, Proposition 1.1] to $b+1\in H^s_0(B_L)$, with source $-g(b)$. It gives a globally C^s representative with zero exterior trace. Since the constructed g is C^2, Lemma A.1 then gives interior classical regularity and the pointwise equation. The radial order, initially almost everywhere, consequently holds pointwise for this continuous representative.

To control the maximum as $L\to\infty$, take $L\ge 2$ and use a smooth competitor equal to 1 in B_{L-1} and to -1 off B_L. Writing it as $2\eta_L-1$, the cutoff can be chosen so that

$$\|\nabla\eta_L\|_2^2+\mathrm{TV}(\eta_L)\le CL^{d-1}, \qquad \|\eta_L\|_2^2\le CL^d.$$

The gradient estimate, bounded variation estimate, and L^2 estimate respectively imply

$$\|\eta_L(\,\cdot+Y)-\eta_L\|_2^2\le C\min\{|Y|^2L^{d-1},\ |Y|L^{d-1},\ L^d\}.$$

Integration over displacements of size below 1, between 1 and L, and above L gives

$$\begin{aligned}K_{d,s}(2\eta_L-1)&\le CL^{d-1}\int_0^1 t^{1-2s}\,\mathrm{d}t+CL^{d-1}\int_1^L t^{-2s}\,\mathrm{d}t+CL^d\int_L^\infty t^{-1-2s}\,\mathrm{d}t\\&\le\begin{cases}CL^{d-2s}, & 0<s<1/2,\\ CL^{d-1}\log L, & s=1/2,\\ CL^{d-1}, & 1/2<s<1.\end{cases}\end{aligned}$$

The potential contribution of the transition shell is $O(L^{d-1})$. Minimality and nonnegativity of the kinetic energy yield

$$\frac{1}{|B_L|}\int_{B_L}G(b)\,\mathrm{d}X\le\begin{cases}CL^{-2s}, & s<1/2,\\ C(\log L)/L, & s=1/2,\\ C/L, & s>1/2.\end{cases}$$

The constants are independent of L for the fixed perturbation δ, and each bound tends to zero. If $b(0) \le 1-\varepsilon$ along a sequence, radial monotonicity gives

$$G(b) \ge \min_{[-1,1-\varepsilon]} G > 0 \quad \text{throughout } B_L,$$

contradicting these bounds. Thus $b(0) \to 1$. For large L the minimizer is nonconstant. At an interior endpoint value the fractional maximum principle and $g(\pm 1) = 0$ would force it to be constant. Thus $-1 < b < 1$ inside, completing (2.16).

Finally, at a global contact X_*, the singular integral gives $L_{d,s}v(X_*) \le L_{d,s}b(X_*)$. If the common value is above α_-, it is in $(\alpha_-, 1)$, and hence

$$0 \le cv_z(X_*) = L_{d,s}v(X_*) + f(v(X_*)) \le f(b(X_*)) - g(b(X_*)) < 0.$$

This contradiction uses the global order. For a translate of b, the order off its support is automatic whenever $v \ge -1$. □

Lemma 2.7 (Entire radial equilibria)**.** *Assume* (A). *Let $d \ge 1$ and $0 < s < 1$. Suppose that $\phi : \mathbb{R}^d \to [-1,1]$ is a bounded radial stationary solution and $\phi(r) \to \beta$ as $r \to \infty$. Then*

$$f(\beta) = 0, \qquad F(\beta) \ge F(\phi(0)).$$

In particular, a radial stationary solution with limit -1 is identically -1.

Proof. Translations of ϕ toward radial infinity converge locally uniformly to β. The interior compactness of Lemma A.1 passes the stationary equation to this constant limit, giving $f(\beta) = 0$. Let $\Phi(r,y)$ be the Poisson extension and put $a = 1-2s$. In the $(1+a)$ conormal convention of Cabré–Sire, its reaction and potential are

$$\widetilde{f} = -\frac{1+a}{d_s} f, \qquad \widetilde{F} = \frac{1+a}{d_s} F.$$

Their regularity assumption is satisfied with any $\max(0,a) < \gamma < 1$: multiply $\widetilde{f}$ by a smooth compactly supported cutoff equal to one near $[-1,1]$ and choose its potential to agree with $\widetilde{F}$ there. The bounded extension still solves the same equation.

The radial Hamiltonian theorem [1, Theorem 2.5], multiplied by $d_s/(1+a)$, states that

$$\mathcal{H}(r) = \frac{d_s}{2} \int_0^\infty y^a (\Phi_r^2 - \Phi_y^2)\, \mathrm{d}y - F(\phi(r))$$

is finite and nonincreasing for $r \ge 0$. This theorem includes $d = 1$. Since the trace has the radial limit β, [1, Lemma 5.1, equation (5.2)] gives $\int_0^\infty y^a |\nabla \Phi(r,y)|^2 \, \mathrm{d}y \to 0$ as $r \to \infty$. Hence $\mathcal{H}(\infty) = -F(\beta)$. Radial symmetry gives $\Phi_r(0,y) = 0$, and monotonicity yields

$$-F(\beta) \le \mathcal{H}(0) = -\frac{d_s}{2} \int_0^\infty y^a \Phi_y(0,y)^2 \, \mathrm{d}y - F(\phi(0)) \le -F(\phi(0)).$$

This proves the asserted inequality. If $\beta = -1$, assumption (A) forces $\phi(0) \in \{-1, 1\}$. A nonconstant continuous function in $[-1,1]$ attaining either endpoint at the origin has a strictly signed fractional Laplacian there, contradicting the stationary equation and $f(\pm 1) = 0$. Thus ϕ is constant, and its limit selects -1. □

3. Ordered cylindrical waves

A stationary layer also rules out a bounded stationary transition with constant lower exterior data. This will determine the lower end of each half-cylinder profile without an energy assumption at its boundary.

3.1. **Excluding a stationary bubble.**

Lemma 3.1 (No bounded stationary bubble)**.** *Assume* (A)*. Let* $d \geq 1$*,* $0 < s < 1$*, and let* $D \subset \mathbb{R}^d$ *be bounded and open. If* $\phi \in C(\mathbb{R}^d) \cap C^2(D)$ *is bounded and satisfies*

$$-1 \leq \phi \leq 1, \qquad \phi = -1 \text{ on } D^c, \qquad L_{d,s}\phi + f(\phi) = 0 \text{ in } D,$$

then $\phi \equiv -1$*.*

Proof. The reference layer of Lemma 2.1 gives $q_t(X) = q(X_1 + t)$ with $L_{d,s}q_t + f(q_t) = 0$, by (2.2). An interior value $\phi = 1$ would have $L_{d,s}\phi > 0$, since $\phi = -1$ outside the bounded set, contradicting $f(1) = 0$. Thus continuity gives $\max \phi < 1$. If $\phi(P_0) > -1$, the shifts dominating ϕ globally form a nonempty upper interval with finite left endpoint t_*: large positive shifts dominate uniformly on $\overline{D}$, whereas large negative shifts fail at P_0. At t_* a compactness argument gives contact in $\overline{D}$; otherwise its positive gap permits a smaller shift. The contact point P lies in D, since $\phi = -1 < q_{t_*}$ on ∂D. The nonnegative difference $v = q_{t_*} - \phi$ therefore satisfies

$$L_{d,s}v(P) = -C_{d,s} \int_{\mathbb{R}^d} \frac{v(Y)}{|P - Y|^{d+2s}} \, \mathrm{d}Y < 0,$$

contradicting the two equations at their common value. Here local C^2 regularity and $\nabla v(P) = 0$ ensure convergence near P, boundedness controls infinity, and $v > 0$ on D^c gives strictness. □

The cylinder construction will preserve its geometric orders at the approximation level. To this end the singular kernel is capped at the origin, so that each transverse slice remains radially decreasing.

3.2. **Positive-kernel approximations.**

Lemma 3.2 (Ordered half-cylinders)**.** *Assume* (A)*. For* $n \geq 1$*,* $0 < s < 1$*,* $c > 0$*, and* $R > 0$*, put* $\Omega_R = B_R^n \times (-\infty, 0)$*. There are bounded classical solutions* W_R *of* (1.2) *in* Ω_R*, with exterior values* 1 *on* $\{z \geq 0\}$ *and* -1 *on* $\{z < 0,\ |x| \geq R\}$*, obtained as limits of the kernels*

$$k_\varepsilon(Y) = C_{n+1,s} \min\{\varepsilon^{-n-1-2s}, |Y|^{-n-1-2s}\}, \qquad L_\varepsilon v = \lambda_\varepsilon v - k_\varepsilon * v, \quad \lambda_\varepsilon = \int_{\mathbb{R}^{n+1}} k_\varepsilon. \tag{3.1}$$

They have radial nonincreasing transverse slices, are nondecreasing in z*, and increase with* R *when* R *ranges over positive integers. For every* $b \in \mathbb{R}^n$ *and integer* $k > |b|$*,*

$$W_{R-k}(x, z) \leq W_R(x + b, z) \leq W_{R+k}(x, z) \qquad (R > k). \tag{3.2}$$

The limits are selected compatibly within each countable family $\mathcal{R}_\theta = \{m + \theta : m = 0, 1, 2, \dots\}$*,* $0 < \theta \leq 1$*: one common sequence* $\varepsilon_j^{(\theta)} \downarrow 0$ *is used for all radii in that family. In particular,* $\theta = 1$ *gives a common sequence for all positive integer radii. They satisfy the smooth-patch boundary estimates of Lemma A.2.*

Proof. Choose $M > \max\{0, \max_{[-1,1]} f'\}$ and write $H(t) = Mt - f(t)$, $\kappa_\varepsilon = M + \lambda_\varepsilon$. For bounded forcing h, the proper linear equation is solved by the fixed point

$$u(x, z) = e^{\kappa_\varepsilon z/c} + \frac{1}{c} \int_z^0 e^{-\kappa_\varepsilon (t-z)/c} [h + k_\varepsilon * \operatorname{Ext} u](x, t) \, \mathrm{d}t, \tag{3.3}$$

where Ext supplies the specified full exterior data. The right side is a contraction in the interior supremum norm, with factor $\lambda_\varepsilon/\kappa_\varepsilon < 1$. Its fixed point has a locally absolutely continuous vertical representative for every interior x, solves the linear equation distributionally, and tends to 1 at $z = 0^-$.

We record the proper comparison used throughout the construction. On a common cylinder $D = B_\rho(x_0) \times (-\infty, T)$, suppose two bounded functions solve the proper linear equation, with ordered forcing in D, ordered full values on D^c, and ordered one-sided top values. Their vertical representatives are locally absolutely continuous. If w is the larger-data function minus the other, then integration of the vertical equation and positivity of the kernel give

$$w(x, z) \geq -\frac{\lambda_\varepsilon}{\kappa_\varepsilon} (1 - e^{-\kappa_\varepsilon (T-z)/c}) \|w_-\|_{L^\infty(D)}.$$

Taking negative suprema and using $\lambda_\varepsilon/\kappa_\varepsilon < 1$ proves $w \geq 0$. The same calculation applies to a supersolution and a subsolution. It uses the vertical endpoint and the full exterior order; no lateral trace is required. In particular the fixed point is increasing in the forcing, exterior data, and top value. When $-M \leq h \leq M$, comparison with the constants ± 1 bounds the fixed point by these constants.

Begin with $u_0 = -1$ inside the cylinder and define u_{m+1} by (3.3) with $h = H(u_m)$. The function H is increasing and $H(\pm 1) = \pm M$. Thus $-1 \leq u_m \leq u_{m+1} \leq 1$. Their pointwise limit $W_{R,\varepsilon}$ satisfies the nonlinear integral formula by dominated convergence. In particular, it solves $L_\varepsilon W - cW_z + f(W) = 0$ distributionally and retains the vertical regularity and endpoint value of the formula.

Radial decrease follows directly from positive convolution. In the variable $v = u + 1$, the exterior is zero on the lateral side and 2 above the top, and the formula becomes

$$v(x,z) = 2e^{\kappa_\varepsilon z/c} + \frac{1}{c}\int_z^0 e^{-\kappa_\varepsilon(t-z)/c}[h + M + k_\varepsilon * \operatorname{Ext} v](x,t)\,\mathrm{d}t.$$

The shifted forcing $H(u_m) + M$ is nonnegative and radially nonincreasing whenever u_m is. For fixed vertical displacement, the kernel in the transverse variables is nonnegative and radially nonincreasing. Convolution preserves this class: by layer cake the assertion reduces to monotonicity of the intersection volume of two balls as their centers separate. Bounded nonintegrable slices are first restricted to expanding balls and then recovered by monotone convergence. The contraction iteration starting at zero for v, restriction to B_R, and the outer iteration consequently preserve the asserted radial order.

The initial full function u_0 is vertically nondecreasing by its definition. If this holds for u_m, the increasing function H orders the forcing for $u_{m+1}(x, z + a)$ and $u_{m+1}(x, z)$ on $B_R \times (-\infty, -a)$, for $a > 0$. The shifted function has top trace 1, while the other top value is at most 1. Above this smaller cylinder the shifted full function is 1 and dominates the unshifted one; on its lateral exterior below the top both are -1. Proper comparison proves the order for u_{m+1} and hence vertical nondecrease of $W_{R,\varepsilon}$. The same induction on a smaller cylinder proves the radius order: on its lateral exterior the larger-cylinder function is at least -1, and both top values are 1. Center translation commutes with the two iterations; the inclusions $B_{R-k}(0) \subset B_R(-b) \subset B_{R+k}(0)$ then give the analogue of (3.2) before passage to the limit.

Compactness follows from these orders without an estimate uniform in ε for the differential equation. On $B_A^n \times (-T, T)$ the transverse variation of each slice is at most $2|\mathbb{S}^{n-1}|A^{n-1}$ for $n \geq 2$, and at most 4 for $n = 1$. The integrated vertical variation is at most $2|B_A^n|$. The functions are therefore uniformly bounded in BV on every such product, including products meeting the exterior jumps. First choose a sequence $\varepsilon_j \downarrow 0$ along which the local L^1 and almost-everywhere convergences hold simultaneously for every positive integer radius. For each $0 < \theta < 1$, choose a further subsequence $\varepsilon_j^{(\theta)}$ of this sequence along which they hold simultaneously for all radii in the countable family $\mathcal{R}_\theta = \{m + \theta : m = 0, 1, 2, \ldots\}$. Put $\varepsilon_j^{(1)} = \varepsilon_j$. Every $R > 0$ belongs to exactly one such family; define W_R using that family's selected limit. The integer-radius limits are unchanged by the further selections. No common subsequence for all positive real radii is asserted.

The difference of the full and capped kernels is even, nonnegative, and supported in $|Y| < \varepsilon$. Symmetric second differences give

$$|(L_{d,s} - L_\varepsilon)\psi(X)| \leq C\varepsilon^{2-2s} \sup_{B_\varepsilon(X)} |D^2\psi|, \qquad d = n + 1, \tag{3.4}$$

for bounded data that are C^2 near this ball. For compact smooth tests the integrated version is $\|(L_{d,s} - L_\varepsilon)\psi\|_1 \leq C\varepsilon^{2-2s}\|D^2\psi\|_1$. Since $L_\varepsilon\psi \in L^1$, also $L_{d,s}\psi \in L^1$. For $\psi \in C_c^\infty(\Omega_R)$, the

fractional term passes as follows, along the selected sequence:

$$\left|\int_{\mathbb{R}^d} W_{R,\varepsilon}L_\varepsilon\psi - \int_{\mathbb{R}^d} W_R L_{d,s}\psi\right| \le \|W_{R,\varepsilon}\|_\infty \|(L_\varepsilon - L_{d,s})\psi\|_1 + \left|\int_{\mathbb{R}^d} (W_{R,\varepsilon} - W_R)L_{d,s}\psi\right| \longrightarrow 0.$$

The first term uses consistency and the second uses dominated convergence against the fixed integrable function $L_{d,s}\psi$. Local L^1 convergence passes the drift, and boundedness and almost-everywhere convergence pass the reaction. The limit thus solves the full equation in distributions and is classical by Lemma A.1.

The proper forcing $h_\varepsilon = MW_{R,\varepsilon} - f(W_{R,\varepsilon})$ is uniformly bounded. These approximations use exactly the capped kernels, vertical representatives, and assigned top values of Lemma A.2. That lemma applies to the selected limit and gives the asserted boundary powers wherever the exterior data are locally constant. In particular it gives continuity across the top for $|x| < R$ and across the side for $z < 0$, with side-patch constants uniform under downward translation below height -1 for fixed R. No estimate at the corner is required.

Exterior values and the geometric orders pass almost everywhere. Fix $R > k$, $b \in \mathbb{R}^n$, and an integer $k > |b|$. The radii $R - k$, R, and $R + k$ belong to the same family $\mathcal{R}_\theta$, so their capped sandwich passes along its common subsequence in L^1_{loc}; translation by the fixed b preserves this convergence. No additional extraction depends on b. Thus the sandwich holds almost everywhere for every such choice. The interior and lateral continuity just established upgrade it to a pointwise inequality at every $z < 0$; at $z \ge 0$ all profiles equal 1. The integer-radius order passes along the original common sequence. The radial and vertical orders likewise hold for the classical representatives, with the prescribed exterior values. □

3.3. **Exhaustion and the pure phases.**

Proof of Theorem 1.1. We first move the prescribed top of the half-cylinder to infinity. The strict integral maximum principle gives $-1 < W_R < 1$ in Ω_R. By vertical monotonicity the bottom limit $\phi_R(x) = \lim_{z\to-\infty} W_R(x,z)$ exists. Downward translations and Lemma A.1 show that it is a stationary solution in B_R; the vertical coordinate is integrated out using (2.2). It equals -1 outside the ball. The side estimate of Lemma A.2, uniform below height -1, gives its continuous trace across the sphere. Thus Lemma 3.1 implies $\phi_R = -1$. The top trace now permits a choice $a_R > 0$ with $W_R(0, -a_R) = \alpha$.

As integer R tends to infinity, the monotone radius limit is invariant under every transverse translation by (3.2). It has the form $w(z)$, is nondecreasing, equals 1 on $[0,\infty)$, and solves $L_{1,s}w - cw' + f(w) = 0$ on $(-\infty, 0)$. For a fixed positive integer R_0, domination of W_{R_0} gives its continuous top trace and

$$0 \le 1 - w(z) \le 1 - W_{R_0}(0,z) \le C(-z)^p \qquad (z \uparrow 0), \qquad 0 < p < s.$$

The boundary-modulus verification following Lemma 2.4 applies to this profile: the displayed power bound supplies the top estimate, the equation has bounded forcing, and (A) supplies $f \in C^2$ and $f(1) = 0$. Thus the modulus condition holds at every order, and Lemma 2.4 yields $w = 1$. Fix $T > 0$. Since $W_R(0,-T) \to w(-T) = 1 > \alpha$, for all sufficiently large integer R one has $W_R(0,-T) > \alpha$. Together with $W_R(0,-a_R) = \alpha$ and vertical monotonicity, this implies $a_R > T$. Hence $a_R \to \infty$.

Letting R tend to infinity through positive integers, set $U_R(x,z) = W_R(x, z - a_R)$. The domains $B_R \times (-\infty, a_R)$ exhaust $\mathbb{R}^{n+1}$. Lemma A.1, including passage of the operator to the limit, gives a locally C^2 convergent subsequence with limit U, satisfying (1.2), the weak geometric orders and $U(0,0) = \alpha$. An attained value ± 1 would force a constant by the strict singular-integral maximum principle, contrary to the normalization. Hence $-1 < U < 1$.

The three phase limits must still be identified. Vertical monotonicity gives pointwise limits $\phi^\pm(x) = U(x, \pm\infty)$. Interior compactness of the vertical translates identifies these limits as continuous stationary entire radial profiles; uniqueness of the pointwise limit makes the convergence

locally uniform in x. The stationary equations follow by (2.2). Radial monotonicity also gives their radial limits $\beta_\pm$. Lemma 2.7 gives

$$f(\beta_\pm) = 0, \qquad F(\beta_\pm) \geq F(\phi^\pm(0)).$$

We use the same compact-support contact argument in the three phase comparisons. For a profile b from Lemma 2.6 with support radius L, let $v_\tau \geq -1$ globally and suppose $(X,\tau) \mapsto v_\tau(X)$ is continuous on $\overline{B}_L$ times a compact parameter interval. Then

$$m(\tau) = \min_{X \in \overline{B}_L} \{v_\tau(X) - b(X)\}$$

is continuous. Opposite signs at two parameters give an intermediate parameter with $m = 0$. At that parameter $v_\tau \geq b$ globally, because $b = -1$ outside B_L. If $v_\tau > -1$ on ∂B_L, the contact lies inside the ball. The contact calculation in Lemma 2.6 rules it out whenever the common value exceeds α_-, $(v_\tau)_z \geq 0$, and the profile equation holds near the contact point. Only that local equation is used, so this argument also applies to the half-cylinder profiles. No monotonicity in the sliding parameter is required.

Suppose first that ϕ^+ is not identically 1. The strict integral maximum principle already used above excludes $\phi^+(0) = 1$, so its central maximum is $a_0 = \phi^+(0) \in [\alpha, 1)$. The endpoint β_+ cannot be 1, since it is at most a_0; it cannot be -1, since $F(a_0) > 0$. It is therefore an interior root, with $\alpha_- < \beta_+ < \alpha$. Choose the compact comparison profile of Lemma 2.6 in dimension $n+1$, with support radius L and $b(0) > a_0$. Take $\delta > 0$ so small that $\beta_+ - 2\delta > \alpha_-$. Since $\phi^+(x) \geq \beta_+$, local uniform convergence at the upper end gives a high fixed z_0 such that $U(x, z+z_0) > \beta_+ - \delta$ on $\overline{B}_L$. Also $U(0, z_0) \leq a_0 < b(0)$. For a fixed sufficiently large integer $R > L$, local convergence gives $U_R(x, z+z_0) > \beta_+ - 2\delta$ on the same support and $U_R(0, z_0) < b(0)$. Apply the contact argument to $v_\tau(x,z) = U_R(x, z+\tau)$. Here $m(z_0) < 0$, while $m(\tau) > 0$ for sufficiently large τ, when the whole support lies above the artificial top $z = a_R$ of U_R. Continuity on the support through this top follows from its continuous top trace, since $R > L$. At the resulting contact $\tau > z_0$, vertical monotonicity keeps the common value above $\beta_+ - 2\delta > \alpha_-$. The boundary of the barrier has strict order because $|x| \leq L < R$ and $U_R > -1$ there at every height. Contact cannot occur on or above the artificial top, where $U_R = 1 > \max b$, and hence lies in the equation domain. The contact contradiction proves $\phi^+ = 1$.

For the lower profile, $\beta_- \leq \phi^-(0) \leq \alpha$. If $\beta_- \neq -1$, it is an interior root and $U \geq \beta_- > \alpha_-$ globally. Choose a compact comparison profile with $b(0) > \alpha$. For $v_\tau(x,z) = U(x, z+\tau)$, the upper phase just proved gives $m(\tau) > 0$ at a sufficiently high shift, while $m(0) < 0$ at the center. The common contact argument applies, and the contact value is at least $\beta_- > \alpha_-$, a contradiction. Thus $\beta_- = -1$ and Lemma 2.7 gives $\phi^- = -1$.

Radial monotonicity gives a lateral limit $\ell(z) = \lim_{r\to\infty} U(r,z)$. For $r_j \to \infty$, the physical translates $U_j(X,z) = U(|r_j e_1 + X|, z)$ are locally precompact in C^2. For $|X| \leq A$ and $r_j > A$, radial monotonicity gives

$$U(r_j + A, z) \leq U_j(X,z) \leq U(r_j - A, z).$$

Both bounds tend to $\ell(z)$, so every translated limit is independent of X and equals $\ell(z)$. Passing the equation to this limit and using (2.2) gives

$$L_{1,s}\ell - c\ell' + f(\ell) = 0 \quad \text{on } \mathbb{R}.$$

The global derivative bounds pass to ℓ, which is also nondecreasing. Since $-1 \leq \ell(z) \leq U(0,z)$, its lower endpoint is -1. If ℓ is nonconstant, Lemma 2.5 says that its upper endpoint is a root distinct from both endpoints ± 1: -1 would make it constant by monotonicity, and 1 would make $\int (\ell')^2 = 0$ by balance. Denote this root by $\beta \in (\alpha_-, \alpha)$. Choose z_0 so that $\ell(z) > \alpha_-$ for $z \geq z_0$, and a compact comparison profile with $b(0) > \beta$ and radius L. At a sufficiently high fixed height $z_* > z_0 + L$, local uniform convergence to the upper phase gives $m(0) > 0$ for the family $v_\tau(x,z) = U(x + \tau e_1, z + z_*)$, $\tau \geq 0$. For large τ, its center value tends to $\ell(z_*) \leq \beta < b(0)$, so $m(\tau) < 0$. The common contact argument therefore applies. Every point in the translated support has height greater than z_0, where $U(x,z) \geq \ell(z) > \alpha_-$. This is again a contradiction. It follows that $\ell = -1$. The vertical limits are locally uniform in x as established above. The

lateral limit is locally uniform in z by Dini's theorem on compact height intervals, applied to the monotone continuous profiles.

It remains to make the orders strict. Lemma A.1 gives a global $C^{2,\eta}$ bound for U, with $\max(0, 2s-1) < \eta < \min(2s, 1)$; this interval is nonempty. Every first derivative v satisfies the differentiated equation in distributions. Its symmetric second differences are $O(|Y|^{1+\eta})$ with $1+\eta > 2s$, so its fractional Laplacian is continuous and the equation holds pointwise:

$$L_{n+1,s}v - cv_z + f'(U)v = 0.$$

For the bounded function $v = U_z \geq 0$, at a zero minimum the gradient vanishes, and the fractional Laplacian is strictly negative unless $v \equiv 0$. The two vertical phases exclude that alternative. Thus $U_z > 0$. The same linearized equation holds for $v = -U_{x_1}$. It is antisymmetric across $x_1 = 0$ and nonnegative when $x_1 > 0$. At a zero P in this half-space, the equation would give $Lv(P) = 0$. Writing Y^* for reflection in the hyperplane, the integral instead has the form

$$L_{n+1,s}v(P) = C_{n+1,s}\int_{Y_1>0} v(Y)\left(\frac{1}{|P-Y^*|^{n+1+2s}} - \frac{1}{|P-Y|^{n+1+2s}}\right)\mathrm{d}Y < 0$$

unless $v \equiv 0$. The kernel difference is strictly negative. The identically zero case and radial symmetry would make U independent of every transverse variable, contradicting the lateral and upper phase limits. Hence $U_r < 0$ for $r > 0$.

Both vertical phases are attained as limits at every radius, so each $\theta \in (-1, 1)$ defines a unique level height. The implicit function theorem applied to the even Cartesian restriction $(r, z) \mapsto U(re_1, z)$ gives an even C^2 function H_θ of signed r. Its restriction to $r \geq 0$ is the required radial graph, with $H'_\theta(0) = 0$ and $H'_\theta = -U_r/U_z > 0$ for $r > 0$. If $H_\theta(r_j) \leq Z$ along some $r_j \to \infty$, vertical monotonicity would give $U(r_j, Z) \geq \theta$, contradicting $U(r_j, Z) \to -1$. Consequently $H_\theta(r) \to \infty$. □

4. Planar limits and rigidity

Throughout this section we assume (A), $0 < s < 1$ and $c > 0$, with the stable cuts of (2.3). The preparatory estimates and planar rigidity hold at every order.

4.1. **Decay measured from the transition region.**

Lemma 4.1 (Decay in the stable phases)**.** *Let $d \geq 1$ and let v be an entire bounded classical solution of $L_{d,s}v - cv_z + f(v) = 0$, with $-1 < v < 1$. Suppose that*

$$\Gamma_v = \{\alpha_- \leq v \leq \alpha\} \neq \varnothing, \qquad D_v(X) = \operatorname{dist}(X, \Gamma_v).$$

There is a constant depending only on the equation and its global regularity bounds such that

$$1 - v(X) \leq C(1 + D_v(X))^{-2s} \qquad \text{if } v(X) \geq \alpha, \tag{4.1}$$

$$1 + v(X) \leq C(1 + D_v(X))^{-2s} \qquad \text{if } v(X) \leq \alpha_-, \tag{4.2}$$

$$|\nabla v(X)| + |D^2 v(X)| \leq C(1 + D_v(X))^{-2s} \qquad \text{for all } X. \tag{4.3}$$

All derivatives in this statement are Cartesian.

Proof. Let $A = L_{d,s} - c\partial_z + \lambda_0$, with positive kernel G from Lemma B.1. Since $f(1) = 0$ and (2.3) holds,

$$A(1 - v) = f(v) + \lambda_0(1 - v) \leq C\mathbf{1}_{\{v<\alpha\}}.$$

On $\{v \geq \alpha\}$ the expression on the left is in fact at most $-\lambda_0(1 - v)$. If $v(X) \geq \alpha$, the ball $B_{D_v(X)}(X)$ contains no point with $v < \alpha$, since a segment to such a point would meet the level α sooner. The resolvent comparison and its tail estimate give (4.1). Likewise,

$$A(1 + v) = -f(v) + \lambda_0(1 + v) \leq C\mathbf{1}_{\{v>\alpha_-\}},$$

which proves (4.2).

For the derivatives we use a distributional convexity inequality. Suppose that h, a_0, k_0 are bounded continuous functions and $Ah = a_0h + k_0$ in distributions. Let $h_\delta = \varphi_\delta * h$, where φ_δ is a

nonnegative smooth mollifier of mass one, and let $\Phi_\varepsilon(t) = \sqrt{t^2+\varepsilon^2}$. Convexity in the fractional integral and the ordinary chain rule for the drift give

$$\begin{aligned} A\Phi_\varepsilon(h_\delta) &\le \Phi_\varepsilon'(h_\delta)(\varphi_\delta * (a_0 h + k_0)) \\ &\quad + \lambda_0(\Phi_\varepsilon(h_\delta) - h_\delta \Phi_\varepsilon'(h_\delta)). \end{aligned}$$

Test this inequality against a nonnegative $\psi \in C_c^\infty(\mathbb{R}^d)$. The adjoint test function $A^*\psi = L_{d,s}\psi + c\psi_z + \lambda_0\psi$ belongs to L^1. For fixed ε, the functions $\Phi_\varepsilon(h_\delta)$ converge pointwise and remain uniformly bounded, while the right-hand side converges locally uniformly. Thus $\delta \downarrow 0$ is legitimate. Next, as $\varepsilon \downarrow 0$,

$$\Phi_\varepsilon'(h)(a_0 h + k_0) \longrightarrow a_0|h| + \operatorname{sgn}_0(h)k_0 \le a_0|h| + |k_0|,$$

where sgn_0 is the sign function with value zero at the origin. The remaining mass term is between 0 and $\lambda_0\varepsilon$. Dominated convergence therefore gives

$$A|h| \le a_0|h| + |k_0| \quad \text{in distributions.} \tag{4.4}$$

This does not require a pointwise formula for $L_{d,s}h$.

For $h = \partial_i v$, the differentiated equation is $Ah = (\lambda_0 - f'(v))h$. The coefficient is at most $-\lambda_0$ off Γ_v, and the derivatives are globally bounded by Lemma A.1. Therefore $A|h| \le C\mathbf{1}_{\Gamma_v}$. Convolution with G and (B.3) give the first-derivative estimate. For $h = \partial_{ij} v$, differentiation in distributions gives

$$Ah = (\lambda_0 - f'(v))h - f''(v)\partial_i v \partial_j v.$$

Both h and this right-hand side are continuous and bounded, so (4.4) applies once more and gives

$$A|h| \le C\mathbf{1}_{\Gamma_v} + C|\nabla v|^2.$$

If $D = D_v(X) \ge 2$, split the convolution of the last term at displacement $D/2$. In the inner part, $D_v(X-Y) \ge D/2$, so its contribution is $O(D^{-4s})$. In the outer part, the global derivative bound and the kernel tail give $O(D^{-2s})$. The convolution of the indicator has the latter bound as well. Boundedness covers $D < 2$. Only two derivatives of f have entered this argument. □

4.2. **A controlled planar transition strip.**

Lemma 4.2 (The core of a doubly monotone profile)**.** *Assume* (A)*,* $0 < s < 1$ *and* $c > 0$*. Suppose that* u *is a bounded entire solution of*

$$L_{2,s}u - cu_z + f(u) = 0, \qquad -1 \le u \le 1, \quad u_\xi \le 0, \quad u_z \ge 0, \quad u(0,0) = \alpha.$$

Then $-1 < u < 1$*,* $-u_\xi > 0$ *everywhere,* $u(-\infty, z) = 1$ *locally uniformly in* z*, and*

$$\inf_{\{\alpha_- \le u \le \alpha\}} (-u_\xi) > 0. \tag{4.5}$$

Proof. The normalization and $f(\alpha) < 0$ exclude a constant; the integral maximum principle excludes attained endpoints. The first-derivative argument in the proof of Theorem 1.1 applies to u and all its translated limits, by the global bounds of Lemma A.1. Their linearized equations therefore hold pointwise, and we use the same zero-minimum argument below.

Horizontal monotonicity gives the limit $\varphi_-(z) = u(-\infty, z)$. Compactness of horizontal translates and (2.2) identify it as a nondecreasing one-dimensional solution with bounded derivative. Lemma 2.5 makes its endpoints roots of f. Its upper endpoint is at least $\varphi_-(0) \ge \alpha$; the only root in $[\alpha, 1]$ is 1. The same lemma gives

$$c\int_{\mathbb{R}} (\varphi_-')^2 \, \mathrm{d}z = F(1) - F(\varphi_-(-\infty)) \le 0.$$

Thus $\varphi_- = 1$; Dini's theorem and horizontal order make the convergence uniform on compact height intervals.

If (4.5) fails, translate core points where $-u_\xi \to 0$: write $u_j(\xi, z) = u(\xi + \xi_j, z + z_j)$. Compactness gives a limit w with $w(0,0) \in [\alpha_-, \alpha]$ and $w_\xi(0,0) = 0$. The zero-minimum argument for $-w_\xi \ge 0$ gives $w_\xi \equiv 0$. Write $w = \varphi(z)$ and $b = \varphi(+\infty)$. The speed identity and the stable cuts imply $\alpha_- < b < \alpha$: b is a root above the nonroot α_-, and $b = 1$ would force $\varphi \equiv 1$.

Take the compact profile b_0 of Lemma 2.6, with support radius ℓ relative to -1 and $b_0(0) > \alpha$. Choose $T, \delta > 0$ so that $\varphi > \alpha_- + 2\delta$ on $[T-\ell, T+\ell]$. Fix j sufficiently large that

$$u_j > \alpha_- + \delta \text{ on } \overline{B}_\ell((0,T)), \qquad u_j(0,T) < b_0(0).$$

All subsequent horizontal slides use this fixed profile. Its left-state convergence is uniform on the compact height interval $[T-\ell, T+\ell]$; no uniformity in j is required. The continuous nonincreasing function

$$m(a) = \min_{(\eta,t)\in\overline{B}_\ell} \{u_j(a+\eta, T+t) - b_0(\eta,t)\}$$

is positive at some finite $a_0 < 0$ and negative at 0. The compact-support contact argument in the proof of Theorem 1.1 gives an interior contact with the barrier centered at (a_*, T) for some $a_* \in (a_0, 0)$. Horizontal order keeps the common value above $\alpha_- + \delta$, so Lemma 2.6 contradicts $u_{j,z} \geq 0$ there. This proves the uniform core bound. The same zero-minimum argument for $-u_\xi$ now gives its positivity everywhere. □

4.3. **Derivative-cone rigidity.**

Lemma 4.3 (Planar rigidity)**.** *Assume* (A)*,* $0 < s < 1$ *and* $c > 0$*, and normalize the stationary layer by* $q(0) = \alpha$*. If a bounded entire solution* u *satisfies*

$$L_{2,s}u - cu_z + f(u) = 0, \qquad -1 \leq u \leq 1, \quad u_\xi \leq 0, \quad u_z \geq 0, \quad u(0,0) = \alpha, \tag{4.6}$$

then $u(\xi, z) = q(-\xi)$ *throughout* $\mathbb{R}^2$*.*

Proof. Lemma 4.2 gives $p = -u_\xi > 0$, $v = u_z \geq 0$, and $\kappa = \inf_{\mathcal{C}} p > 0$ on $\mathcal{C} = \{\alpha_- \leq u \leq \alpha\}$. The differentiated equations and the zero-minimum argument used in that lemma apply also to constant linear combinations of the first derivatives. For $K \geq 0$ put $W_K = Kp - v$. If $W_K \geq 0$ on $\mathcal{C}$, then $W_K \geq 0$ everywhere. Indeed,

$$L_{2,s}W_K - c(W_K)_z + f'(u)W_K = 0.$$

If $m = \inf W_K < 0$, choose X_j with $W_K(X_j) < 0$ and $W_K(X_j) \to m$. After extraction, $u_j = u(\cdot + X_j)$ converges locally in C^2 to u_*. The corresponding functions $W_{K,j} = K(-u_{j,\xi}) - u_{j,z}$ are bounded below by m everywhere; hence their limit has a global minimum $W_{K,*}(0) = m$. Since $W_K \geq 0$ on the core, $u_j(0) \in [-1, \alpha_-] \cup [\alpha, 1]$. This closed-set condition passes to the limit and gives $f'(u_*(0)) \geq 2\lambda_0$. The drift vanishes and the fractional Laplacian is nonpositive, so the displayed equation gives $0 \leq f'(u_*(0))m < 0$, a contradiction.

The admissible coefficients form a nonempty set because $p \geq \kappa$ on the core and v is bounded; for instance $K = \|v\|_\infty/\kappa$ works there and hence everywhere. The set is closed, since its pointwise inequalities are preserved under limits of coefficients. Consequently its least coefficient

$$K_* = \inf\{K \geq 0 : Kp - v \geq 0 \text{ on } \mathbb{R}^2\}$$

is attained. Suppose for contradiction that $K_* > 0$. If $\delta := \inf_{\mathcal{C}}(K_*p - v) > 0$, choose

$$0 < \varepsilon < \min\{K_*/2, \delta/(2\|p\|_\infty)\}.$$

Then $(K_* - \varepsilon)p - v \geq \delta/2$ on the core and hence is nonnegative everywhere, a contradiction. Thus $\inf_{\mathcal{C}}(K_*p - v) = 0$. Translate core points approaching zero. The limit satisfies $\widehat{W} \geq 0$, $\widehat{W}(0) = 0$ and $-\widehat{u}_\xi(0) \geq \kappa$. The zero-minimum argument gives $\widehat{W} \equiv 0$, or $\widehat{u}_z + K_*\widehat{u}_\xi = 0$. Solving along the straight characteristics yields $\widehat{u}(\xi, z) = \phi(-\xi + K_*z)$, where ϕ is nonconstant and nondecreasing. Rotation invariance, (2.2), and one-dimensional scaling give

$$(1 + K_*^2)^s L_{1,s}\phi - cK_*\phi' + f(\phi) = 0.$$

Put $a = (1 + K_*^2)^s$. Division by a gives the equation in Lemma 2.5, with positive speed cK_*/a, reaction f/a, and potential F/a. Its derivative is bounded by the inherited global regularity. Thus its endpoint limits are roots of f, and multiplication of the speed identity by a gives

$$cK_* \int_{\mathbb{R}} (\phi')^2 \, \mathrm{d}t = F(\phi(+\infty)) - F(\phi(-\infty)).$$

Moreover, $\phi'(0) \geq \kappa$ and $\phi(0) \in [\alpha_-, \alpha]$, so ϕ takes values in (α_-, α) even if $\phi(0)$ is one of the cut levels. For any fixed (ξ, z) with $\alpha_- < \widehat{u}(\xi, z) < \alpha$, the translated solutions $u_j \to \widehat{u}$ satisfy $\alpha_- < u_j(\xi, z) < \alpha$ for all large j. The original core bound is translation invariant, so $-(u_j)_\xi(\xi, z) \geq \kappa$ there; local C^1 convergence gives $-\widehat{u}_\xi(\xi, z) \geq \kappa$. Since $\widehat{u}(\xi, z) = \phi(-\xi + K_* z)$, this proves $\phi' \geq \kappa$ wherever $\alpha_- < \phi < \alpha$. If either endpoint were an interior root, it would lie in (α_-, α). Convergence would then place ϕ in this interval on an entire infinite half-line, where integrating $\phi' \geq \kappa$ contradicts boundedness. The cut levels are not roots, so the only possible endpoints are ± 1. They are distinct because $\phi'(0) \geq \kappa > 0$; monotonicity therefore orders them as $-1, 1$. The displayed speed identity now has zero right-hand side and strictly positive left-hand side, a contradiction. Thus $K_* = 0$ and $u_z = 0$. Write the stationary monotone profile as $u(\xi) = \widetilde{q}(-\xi)$. Its upper endpoint is 1 by Lemma 4.2. Translations toward its lower endpoint pass the stationary equation to a constant, so that endpoint is a root. The same endpoint-exclusion argument, using the original core bound, forces it to be -1. Lemma 2.1 and $\widetilde{q}(0) = \alpha$ give $\widetilde{q} = q$. □

4.4. **The cylindrical layer limit.**

Lemma 4.4 (The profile at large radius)**.** *Assume* (A)*, $n \geq 1$, $0 < s < 1$ and $c > 0$. Let $U = U(r, z)$ be a bounded classical solution of* (1.2) *with*

$$-1 < U < 1, \qquad U_z > 0, \qquad U_r < 0 \quad (r > 0),$$

and the three pure-phase limits in (1.4)*, locally uniform in the remaining variables. For all large z, the radius $R(z)$ defined by $U(R(z), z) = \alpha$ satisfies*

$$R \to \infty, \qquad R' > 0, \qquad R' \to 0, \qquad R = o(z).$$

The physical wave has bounded global $C^{2,\eta}$ norm for every $0 < \eta < \min\{2s, 1\}$, and bounded global $C^{3,\gamma}$ norm for $0 < \gamma < 2s - 1$ when $s > 1/2$. With $q(0) = \alpha$,

$$U(R(z) + \xi, z + \tau) \longrightarrow q(-\xi) \quad \text{locally in } C^2(\mathbb{R}^2). \tag{4.7}$$

Equivalently, the physical translates $U(|R(z)e_1 + X|, z + T)$ converge locally in $C^2(\mathbb{R}^{n+1})$ to $q(-X_1)$. For all large z and every $r \geq 0$,

$$|U(r, z) - \operatorname{sgn}(R(z) - r)| + |\nabla U(r, z)| + |D^2 U(r, z)| \leq C(1 + |r - R(z)|)^{-2s}. \tag{4.8}$$

All derivatives are Cartesian derivatives of the physical wave. For $\Gamma = \{\alpha_- \leq U \leq \alpha\}$ there are constants $c_0, C_0 > 0$ such that at all large heights

$$\operatorname{dist}((x, z), \Gamma) \geq c_0(|r - R(z)| - C_0)_+. \tag{4.9}$$

For every fixed $\theta \in (-1, 1)$, the level set is an even C^2 graph $z = H_\theta(r)$ in signed r, with $H'_\theta(r) > 0$ for $r > 0$ and $H_\theta(r) \to \infty$ as $r \to \infty$. Its positive inverse radius R_θ is defined for $z > H_\theta(0)$ and satisfies

$$R_\theta(H_\theta(r)) = r \quad (r > 0). \tag{4.10}$$

Moreover,

$$R_\theta(z) - R(z) \longrightarrow -q^{-1}(\theta). \tag{4.11}$$

Proof. Lemma A.1 gives the stated global regularity. The level-graph argument at the end of the proof of Theorem 1.1 uses only the present orders and phase limits. It gives the asserted properties of H_θ; strict increase and divergence make R_θ its inverse on $(H_\theta(0), \infty)$, proving (4.10). For every fixed $A \geq 0$, the upper phase gives $U(A, z) > \alpha$ at sufficiently large heights, while the lateral limit is $-1 < \alpha$. Continuity and strict radial decrease give a unique radius $R(z) > A$ there. This proves both its large-height existence and $R(z) \to \infty$. The implicit function theorem gives $R' = -U_z/U_r > 0$. For any $z_j \to \infty$, put $R_j = R(z_j)$ and translate the physical wave:

$$U_j(X, t) = U(|R_j e_1 + X|, z_j + t).$$

Compactness gives a locally C^2 convergent subsequence. The expansion $|R_j e_1 + X| - R_j = X_1 + O(|X|^2/R_j)$ and the uniform first derivative bound make its limit independent of $X_2, \ldots, X_n$. Local second differences and the bounded far kernel pass the equation to the limit; normalized

dimension reduction gives a planar solution Ψ with $\Psi(0,0)=\alpha$, $\Psi_\xi \le 0$, $\Psi_t \ge 0$; on fixed compact sets the X_1 derivative of $|R_j e_1 + X|$ is eventually positive. Lemma 4.3 identifies it as $q(-\xi)$. The limit is independent of the subsequence, proving (4.7). Consequently

$$U_z(R(z),z) \to 0, \qquad U_r(R(z),z) \to -q'(0) < 0,$$

so $R' \to 0$. Given $\varepsilon > 0$, choose Z_ε so that $0 < R' \le \varepsilon$ above it. Then

$$R(z) \le R(Z_\varepsilon) + \varepsilon(z - Z_\varepsilon), \qquad z \ge Z_\varepsilon,$$

and division by z followed by $\varepsilon \downarrow 0$ gives $R = o(z)$. For a fixed $\theta \in (-1,1)$ put $a_\theta = -q^{-1}(\theta)$. The strict monotonicity of q and (4.7) give, for every $\varepsilon > 0$ and all sufficiently large z,

$$U(R(z) + a_\theta - \varepsilon, z) > \theta > U(R(z) + a_\theta + \varepsilon, z).$$

Both radii are positive because $R \to \infty$. Radial order therefore gives a unique $R_\theta(z)$ between them, proving (4.11).

Choose Z_0 so that R is L-Lipschitz above Z_0 and the core $\Gamma = \{\alpha_- \le U \le \alpha\}$ lies there between $r = R(z)$ and $r = R(z) + W$ for a fixed W. For a recent core point (x', z'), $z' \ge Z_0$,

$$|r - R(z)| \le |x - x'| + L|z - z'| + W.$$

By Cauchy–Schwarz, its distance from (x,z) is at least $(|r - R(z)| - W)_+/\sqrt{1+L^2}$. For the older core, choose B so that $U(B, Z_0) < \alpha_-$. Radial and vertical order then give $U(r', z') < \alpha_-$ whenever $r' \ge B$ and $z' < Z_0$; hence every such core point has $|x'| \le B$. At large z, $z - Z_0 \ge R(z) \ge B$ since $R = o(z)$. If $r \le 2R(z)$, its vertical distance is at least $R(z) \ge |r - R(z)|$; if $r \ge 2R(z)$, its transverse distance is at least $r - B \ge r - R(z)$. Taking the infimum over both parts of the core proves (4.9), with $c_0 = (1+L^2)^{-1/2}$ and $C_0 = W$. Lemma 4.1 now gives the phase and Cartesian derivative bounds in (4.8). The upper estimate applies for $r \le R$, the lower for $r \ge R_{\alpha_-}$, and the remaining strip has bounded width by (4.11). □

5. Stress and interface growth

Throughout this section we assume (A), $n \ge 2$, $1/2 < s < 1$ and $c > 0$. Let U be any bounded classical solution of (1.2) with the monotonicity and locally uniform phase limits in (1.3)–(1.4). Let $R(z) = R_\alpha(z)$, so that $U(R(z), z) = \alpha$. Lemma 4.4 gives $R \to \infty$, $R' \to 0$, $R = o(z)$, the local layer limit and the uniform Cartesian derivative tails. All estimates below apply above a fixed sufficiently large height Z.

5.1. **A slice estimate inside a pure phase.**

Lemma 5.1. *Let $d \ge 1$, $0 < s < 1$, $D \ge 1$, and let w be bounded and measurable with $\|w\|_\infty \le K$. If w is C^2 in $B_{4D}(X_0)$ and $\|w\|_{C^2(B_{4D}(X_0))} \le \delta \le K$, then its Poisson extension satisfies*

$$\int_0^\infty y^{1-2s} |\nabla (P_y^{(d)} * w)(X_0, y)|^2 \, dy \le C(d,s)(\delta^2 + K^2 D^{-2s}). \tag{5.1}$$

Here ∇ includes both the physical and extension variables.

Proof. Take $X_0 = 0$ and split $w = \chi w + (1-\chi)w = w_0 + w_1$, where $0 \le \chi \le 1$, $\chi = 1$ in B_{2D}, $\chi = 0$ outside B_{3D} and $|D^j \chi| \le C_j D^{-j}$ for $j \le 2$. The global C^2 norm of w_0 is at most $C\delta$. The cancellation argument for (2.7) gives, for $0 < y \le 1$, tangential derivative bound $C\delta$ and normal derivative bound $C\delta y^{2s-1}$. For $y \ge 1$ both are bounded by $C\delta/y$. Their weighted square integral is bounded by

$$C\delta^2 \left\{ \int_0^1 (y^{1-2s} + y^{2s-1}) \, dy + \int_1^\infty y^{-1-2s} \, dy \right\}.$$

To estimate the remote part, put $v(X) = w_1(DX)$ and $W_v = P_\eta^{(d)} * v$. Here $v = 0$ in B_2 and $|v| \le K$. For $|H| \ge 2$ and $0 < \eta \le 1$,

$$|\nabla_X P_\eta^{(d)}(H)| \le C\eta^{2s} |H|^{-d-2s-1}, \qquad |\partial_\eta P_\eta^{(d)}(H)| \le C\eta^{2s-1} |H|^{-d-2s}.$$

Integrating in H bounds the derivatives of $W_v(0,\eta)$ by

$$CK\eta^{2s}\int_2^\infty t^{-2s-2}\,\mathrm{d}t \quad\text{and}\quad CK\eta^{2s-1}\int_2^\infty t^{-2s-1}\,\mathrm{d}t,$$

respectively. For $\eta\ge 1$ the bound is CK/η. Thus

$$\int_0^\infty \eta^{1-2s}|\nabla W_v(0,\eta)|^2\,\mathrm{d}\eta \le CK^2\left\{\int_0^1(\eta^{1+2s}+\eta^{2s-1})\,\mathrm{d}\eta+\int_1^\infty \eta^{-1-2s}\,\mathrm{d}\eta\right\}<\infty.$$

The extension of w_1 is $W_v(X/D,y/D)$; the change of variable $y=D\eta$ multiplies this energy by D^{-2s}. The inequality $|\nabla W_0+\nabla W_1|^2\le 2|\nabla W_0|^2+2|\nabla W_1|^2$ now proves (5.1), where W_j denotes the extension of w_j. □

5.2. **The stress balance outside a moving radius.** Let V be the extension of the physical wave in $\mathbb{R}^{n+1}$. Cylindrical symmetry gives

$$V_{rr}+\frac{n-1}{r}V_r+V_{zz}+y^{-a}(y^aV_y)_y=0,\qquad -d_s\lim_{y\downarrow 0}y^aV_y=cU_z-f(U). \tag{5.2}$$

Define

$$\begin{aligned}A(r,z)&=d_s\int_0^\infty y^aV_r^2\,\mathrm{d}y,\qquad P(r,z)=d_s\int_0^\infty y^aV_rV_z\,\mathrm{d}y,\\ T(r,z)&=\frac{d_s}{2}\int_0^\infty y^a(V_r^2-V_z^2-V_y^2)\,\mathrm{d}y-F(U(r,z)).\end{aligned} \tag{5.3}$$

All three integrals converge by (2.7). We use the interior radius $h(z)=R(z)/2$ and the quantities

$$B_h=\int_h^\infty U_z(-U_r)\,\mathrm{d}r,\qquad I_h=\int_h^\infty\frac{A}{r}\,\mathrm{d}r,\qquad J_h=\int_h^\infty P\,\mathrm{d}r. \tag{5.4}$$

Lemma 5.2. *The quantities in (5.4) satisfy*

$$c\int_b^d B_h(t)\,\mathrm{d}t=(n-1)\int_b^d I_h(t)\,\mathrm{d}t+J_h(d)-J_h(b)+\int_b^d\{-T(h(t),t)+h'(t)P(h(t),t)\}\,\mathrm{d}t \tag{5.5}$$

for $Z\le b<d$. Moreover,

$$|J_h(z)|\le C,\qquad |T(h(z),z)|+|P(h(z),z)|\le CR(z)^{-2s},\qquad I_h(z)=\frac{\sigma_s+o(1)}{R(z)}. \tag{5.6}$$

In particular, with constants independent of b,d,

$$c\int_b^d B_h=(n-1)\int_b^d I_h+O(1)+O\left(\int_b^d R^{-2s}\right). \tag{5.7}$$

Proof. The Poisson kernel preserves vertical order, and the radial-convolution argument in Section 3, applied at each vertical displacement to $U(\cdot,z)+1$, gives $V_z\ge 0$ and $V_r\le 0$. To pass the lateral phase to V, restrict the displacement to a ball and apply bounded convergence there; the Poisson mass outside the ball is uniformly small. Therefore

$$\int_0^\infty(-V_r(r,z,y))\,\mathrm{d}r=V(0,z,y)+1\le 2. \tag{5.8}$$

Together with (2.7), this proves

$$\sup_z\left\{\int_0^\infty A(r,z)\,\mathrm{d}r+\int_0^\infty|P(r,z)|\,\mathrm{d}r\right\}\le C. \tag{5.9}$$

Indeed the resulting y majorant is $Cy^{1-2s}\min(1,y^{-1})$, which is integrable because $1/2<s<1$. In particular, all the cut quantities are finite and satisfy

$$0\le B_h\le 2\|U_z\|_\infty,\qquad 0\le I_h\le C/h,\qquad |J_h|\le C.$$

Multiplication of (5.2) by $d_sy^aV_r$ and integration in y give, in distributions on $r>0$,

$$\partial_rT+\partial_zP=-\frac{n-1}{r}A-cU_zU_r. \tag{5.10}$$

The normal term at $y = 0$ is $U_r(cU_z - f(U))$; its $f(U)U_r$ part cancels the derivative of the potential in T. We justify this calculation and its integrated form together. First integrate with $0 < \eta < y < Y$ over $h(t) < r < L$, $b < t < d$, where $L > \max_{[b,d]} h$. The outward flux along the moving left boundary is $-T + h'P$, with the y integrals temporarily truncated. For fixed η, Y, the right boundary stress tends to zero as $L \to \infty$, locally uniformly in t: the lateral phase passes to the extension and its derivatives by the Poisson formula, and $F(U(L,t)) \to F(-1) = 0$. Thus this radial limit is legitimate.

The upper y flux integrated in r is $O(Y^{-2s})$, uniformly in t, by (5.8) and (2.7). At the lower boundary put $g_\eta = -d_s\eta^a V_y(\cdot,\cdot,\eta)$ and $g_0 = cU_z - f(U)$. The global conormal convergence proved after (2.7) gives $\|g_\eta - g_0\|_\infty \to 0$. For each fixed t and $a_0 \geq 0$,

$$\int_{a_0}^{\infty} (-V_r(r,t,\eta))\,\mathrm{d}r = V(a_0,t,\eta) + 1 \longrightarrow U(a_0,t) + 1 = \int_{a_0}^{\infty} (-U_r(r,t))\,\mathrm{d}r.$$

Local convergence and nonnegativity, together with convergence of these total masses, give $V_r(\cdot,t,\eta) \to U_r(\cdot,t)$ in $L^1([a_0,\infty))$. Taking $a_0 = h(t)$, the lower flux converges since

$$\|g_\eta V_r(\eta) - g_0 U_r\|_{L^1(h(t),\infty)} \leq 2\|g_\eta - g_0\|_\infty + \|g_0\|_\infty \|V_r(\eta) - U_r\|_{L^1(h(t),\infty)} \longrightarrow 0.$$

The masses are at most 2, so dominated convergence also passes the time integral on $[b,d]$. The integrable majorants in (2.7) pass the moving-boundary stresses as $\eta \downarrow 0$, $Y \uparrow \infty$; the majorant used for (5.9) passes I_h and the endpoint fluxes $J_h(b), J_h(d)$. This proves (5.5) for every $b < d$ without differentiating J_h.

At the moving boundary, fix $0 < \gamma < 1/32$. Since $R = o(z)$, the interval $[z - 4\gamma R(z), z + 4\gamma R(z)]$ lies at large heights when z is large; there $|R'| \leq 1$. Hence

$$R(z') - |x'| \geq R(z)/2 - 8\gamma R(z) \geq R(z)/4 \quad \text{in } B_{4\gamma R(z)}(h(z)e_1, z)$$

for all large z. The Cartesian tail estimate of Lemma 4.4 bounds the C^2 norm of $1 - U$ on this ball by $CR(z)^{-2s}$. Apply Lemma 5.1 with $w = 1 - U$, $K = 2$ and $D = \gamma R(z) \geq 1$; for large z its local norm is at most 2. The resulting slice energy is $O(R^{-2s})$. Also $F(U(h(z),z)) = O(R^{-4s})$ since F has a quadratic zero at 1. These bounds give the second estimate in (5.6).

It remains to localize the measure $A(r,z)\,\mathrm{d}r$. The physical translate limit in Lemma 4.4, the integrable Poisson derivative kernels and dimension reduction give

$$V(R(z) + t, z, y) \longrightarrow Q(-t, y), \qquad V_r(R(z) + t, z, y) \longrightarrow -Q_t(-t, y),$$

uniformly for t in compact intervals and y in compact subintervals of $(0,\infty)$. Also

$$\int_0^\infty V_r^2\,\mathrm{d}r \leq C\min(1, y^{-1}).$$

Thus, for $0 < \eta \leq 1 \leq Y$, the weighted energies on $y < \eta$ and $y > Y$ are at most $C\eta^{2-2s}$ and CY^{1-2s}, respectively, uniformly in z. For $\eta \leq y \leq Y$ and $R(z) > K$, the radial variation outside $|r - R(z)| \leq K$ is bounded by

$$1 - V(R(z) - K, z, y) + V(R(z) + K, z, y) + 1.$$

Its limit is $2 - Q(K,y) + Q(-K,y)$. The endpoint limits of q and uniform Poisson tail bounds for $y \in [\eta, Y]$ show that this tends uniformly to zero as $K \to \infty$. The derivative bound converts variation to square integral. First restricting y to $[\eta, Y]$, then using the two height-tail estimates, gives

$$\lim_{K\to\infty} \limsup_{z\to\infty} \int_{\substack{r\geq 0 \\ |r - R(z)| > K}} A(r,z)\,\mathrm{d}r = 0.$$

The same truncation argument and local convergence therefore give

$$\int_0^\infty A(r,z)\,\mathrm{d}r \longrightarrow d_s \int_{\mathbb{R}} \int_0^\infty y^a Q_t^2\,\mathrm{d}y\,\mathrm{d}t = \sigma_s,$$

where the last equality is (2.8). For $R = R(z) > 2K$, the fixed window lies inside $r \geq h(z)$. There $|R/r - 1| \leq K/(R-K)$, and everywhere on the cut $0 \leq R/r \leq 2$. Consequently

$$\left| RI_h - \int_0^\infty A \, \mathrm{d}r \right| \leq \frac{K}{R-K} \int_0^\infty A \, \mathrm{d}r + \int_{\substack{r\geq 0 \\ |r-R|>K}} A \, \mathrm{d}r.$$

Letting $z \to \infty$ and then $K \to \infty$ proves $R(z)I_h(z) \to \sigma_s$. The other assertions now follow from (5.9), boundedness of h', and (5.5). □

5.3. **Integrated drift and the interface laws.**

Lemma 5.3. *For the same wave and cut,*

$$\int_Z^z B_h(t) \, \mathrm{d}t = m_s R(z) + o(R(z)). \tag{5.11}$$

Proof. Set

$$g(v) = q'(q^{-1}(v)), \quad g(\pm 1) = 0, \qquad G(v) = \int_{-1}^{v} g(w) \, \mathrm{d}w.$$

The strict positivity and decay of q' in Lemma 2.1 give $g \in C([-1,1])$ and $g \geq 0$. Thus G is increasing and Lipschitz, with $G(-1) = 0$; the substitution $v = q(t)$ gives

$$G(1) = \int_{-1}^{1} q'(q^{-1}(v)) \, \mathrm{d}v = \int_{\mathbb{R}} (q'(t))^2 \, \mathrm{d}t = m_s.$$

The local layer limit and the uniform phase and derivative tails imply

$$e(z) := \sup_{r\geq 0} | - U_r(r,z) - g(U(r,z))| \longrightarrow 0.$$

To obtain the supremum, use profile convergence on a fixed interface window and then the tails together with $g(\pm 1) = 0$ on its complement. The integrals

$$N(z) = \int_0^\infty G(U(r,z)) \, \mathrm{d}r, \qquad M_0(z) = \int_0^\infty (U(r,z) + 1) \, \mathrm{d}r$$

are finite, with

$$N(z) = m_s R(z) + O(1), \qquad M_0(z) = 2R(z) + O(1). \tag{5.12}$$

Indeed, (4.8) and the Lipschitz bound for G give

$$|U(r,z) + 1 - 2\mathbf{1}_{\{r<R(z)\}}| + |G(U(r,z)) - m_s \mathbf{1}_{\{r<R(z)\}}| \leq C(1 + |r - R(z)|)^{-2s}.$$

This bound has an r integral bounded independently of $R(z)$ because $2s > 1$.

Let $B(z) = \int_0^\infty U_z(-U_r) \, \mathrm{d}r$; the radial variation of U gives $0 \leq B \leq 2\|U_z\|_\infty$. The fundamental theorem of calculus at each r, followed by Tonelli for the nonnegative integrands, gives

$$M_0(d) - M_0(b) = \int_b^d \int_0^\infty U_z(r,t) \, \mathrm{d}r \, \mathrm{d}t,$$
$$N(d) - N(b) = \int_b^d \int_0^\infty g(U(r,t)) U_z(r,t) \, \mathrm{d}r \, \mathrm{d}t.$$

Both differences are finite by (5.12). Their time densities are therefore locally integrable, so M_0, N are locally absolutely continuous. Multiplying $| - U_r - g(U)| \leq e(t)$ by $U_z \geq 0$ and integrating gives

$$\left| \int_b^d B - N(d) + N(b) \right| \leq \sup_{[b,d]} e \, \{M_0(d) - M_0(b)\}. \tag{5.13}$$

The part discarded by the cut has uniformly bounded total mass. Indeed $r \leq h(t)$ implies $R(t) - r \geq r$, and therefore

$$0 \leq \int_b^d (B - B_h) \, \mathrm{d}t \leq C \int_0^\infty (1+r)^{-2s} \int_b^d U_z(r,t) \, \mathrm{d}t \, \mathrm{d}r$$
$$\leq 2C \int_0^\infty (1+r)^{-2s} \, \mathrm{d}r \leq C.$$

Given $\varepsilon > 0$, choose $Z_\varepsilon \geq Z$ so that $e(t) \leq \varepsilon$ for $t \geq Z_\varepsilon$. Apply (5.13) on $[Z_\varepsilon, z]$. The integral over the fixed interval $[Z, Z_\varepsilon]$ is finite by the bound on B, and the cut discards at most the constant just obtained. Equation (5.12) then yields

$$\left| \int_Z^z B_h(t)\,\mathrm{d}t - m_s R(z) \right| \leq 2\varepsilon R(z) + C_\varepsilon \qquad (z \geq Z_\varepsilon).$$

Divide by $R(z) \to \infty$ and let $\varepsilon \downarrow 0$ to obtain (5.11). □

Lemma 5.4 (Inverting a cumulative power balance)**.** *Let $\gamma > 0$ and let R be positive and continuous on $[Z, \infty)$. If*

$$S(z) = \int_Z^z R(t)^{-\gamma}\,\mathrm{d}t \longrightarrow \infty, \qquad R(z)/S(z) \longrightarrow A > 0,$$

then

$$\frac{R(z)^{1+\gamma}}{z} \longrightarrow (1+\gamma)A. \tag{5.14}$$

Proof. For $z > Z$, the defining integral gives

$$(S^{1+\gamma})' = (1+\gamma)(S/R)^\gamma \longrightarrow (1+\gamma)A^{-\gamma}.$$

Integration yields $S(z)^{1+\gamma}/z \to (1+\gamma)A^{-\gamma}$. Multiplication by $(R/S)^{1+\gamma} \to A^{1+\gamma}$ proves (5.14). Only the defining integral is differentiated. □

Proof of Theorem 1.2. Put

$$S(z) = \int_Z^z \frac{\mathrm{d}t}{R(t)}, \qquad k = (n-1)\sigma_s > 0.$$

Since $R = o(z)$, eventually $R(z) \leq z$, and hence $S \to \infty$. Equations (5.7), (5.6) and (5.11), together with $R^{-2s} = o(R^{-1})$, yield

$$cm_s R(z) + o(R(z)) = kS(z) + o(S(z)) + O(1).$$

Here an error $\varepsilon(t)/R(t)$ with $\varepsilon(t) \to 0$ has integral $o(S(z))$: split at a fixed large height and use $S \to \infty$. This applies both to the error in I_h and to R^{-2s}, since $R^{1-2s} \to 0$. Writing the displayed balance as $(cm_s + o(1))R = (k + o(1))S + O(1)$ and dividing by $R \to \infty$ gives $R/S \to k/(cm_s) > 0$. Lemma 5.4 with $\gamma = 1$ gives

$$\frac{R(z)^2}{z} \longrightarrow \frac{2(n-1)\sigma_s}{cm_s}. \tag{5.15}$$

For every fixed $\theta \in (-1, 1)$, (4.11) transfers this limit to R_θ. Substitute $z = H_\theta(r) \to \infty$ using (4.10) and take reciprocals to obtain (1.8). □

6. Sharp planar asymptotics at every order

In this section $n = 1$, $0 < s < 1$, $c > 0$, and U is any bounded classical solution of (1.2) with the orders and locally uniform phase limits in (1.3)–(1.4). Assumption (A) remains in force. We keep $q(0) = \alpha$, write $\beta = 2s$, and let $R = R_\alpha$. Lemma 4.4 supplies the large-height radius, its flatness and relative-level limits, the local layer profile, and the Cartesian phase and derivative bounds for every $0 < s < 1$. The argument uses the whole radial half-line. Its boundary at the axis retains the interaction between the two sides of the front.

6.1. **A sharper derivative bound.** We retain the transition set Γ_v and distance D_v of Lemma 4.1. A bound on variation along coordinate lines improves its first-derivative estimate by one power. The next lemma applies in arbitrary dimension, independently of the planar restriction on this section's wave.

Lemma 6.1 (Decay from coordinate variation)**.** *Let $d \geq 1$ and $\beta = 2s$. Suppose that v satisfies the hypotheses of Lemma 4.1 and, for a Cartesian coordinate X_i,*

$$\sup_{\widehat{X}_i} \int_{\mathbb{R}} |\partial_i v(X_i, \widehat{X}_i)| \, \mathrm{d}X_i \leq V_i < \infty.$$

Then

$$|\partial_i v(X)| \leq C(1 + D_v(X))^{-1-\beta}. \tag{6.1}$$

The constant may also depend on V_i. For every cylindrical wave satisfying (1.3) and (1.4), in any transverse dimension, this estimate holds for every first Cartesian derivative.

Proof. Set $\mathcal{A} = L_{d,s} - c\partial_z + \lambda_0$. For $h = \partial_i v$, differentiation in distributions gives $\mathcal{A}h = (\lambda_0 - f'(v))h$. Apply the established convexity inequality (4.4). Since $f'(v) \geq 2\lambda_0$ off Γ_v, it yields

$$\mathcal{A}|h| \leq K\mathbf{1}_{\Gamma_v}|h|, \quad K = \max\{0, \sup_{[-1,1]} (\lambda_0 - f')\}.$$

Positive resolvent comparison, (B.2), and $|X - Y| \geq D = D_v(X) \geq 1$ for $Y \in \Gamma_v$ imply

$$|h(X)| \leq CK \int_{\Gamma_v} |X - Y|^{-d-\beta} |h(Y)| \, \mathrm{d}Y \leq CKV_i \int_{\mathbb{R}^{d-1}} \max(D, |\eta|)^{-d-\beta} \, \mathrm{d}\eta \leq CD^{-1-\beta}.$$

Here integrate first in Y_i, then split $|\eta|$ at D; for $d = 1$ the remaining integral has its zero-dimensional meaning. Boundedness handles $D < 1$.

For a wave in any transverse dimension, every transverse coordinate line has the form $t \mapsto U(\sqrt{t^2 + |x_\perp|^2}, z)$, even and monotone on each half-line with endpoint -1. Hence its total variation is $2[U(|x_\perp|, z) + 1] \leq 4$, uniformly in the other coordinates. Every vertical line has variation 2 by its two end states. The vertical limits also make Γ_U nonempty. Thus the hypothesis holds for all first derivatives, including U_r after a horizontal rotation. □

Lemma 6.2 (Radial derivative decay)**.** *For every $0 < s < 1$, at all sufficiently large heights and every $r \geq 0$,*

$$|U_r(r, z)| \leq C(1 + |r - R(z)|)^{-1-\beta}. \tag{6.2}$$

Proof. Apply Lemma 6.1 to U and use the core-distance bound (4.9). For large $|r - R(z)|$ this gives the stated decay; the global derivative bound handles the remaining strip. At the axis $U_r = 0$ by evenness. □

6.2. **An integrable primitive for the drift.** The phase mass need not be integrable at low orders. Composing the profile with the layer derivative produces a primitive whose defect from a step function is integrable for every $s > 0$.

Lemma 6.3 (Weighted layer matching)**.** *Define*

$$g(v) = q'(q^{-1}(v)) \quad (-1 < v < 1), \qquad g(\pm 1) = 0, \qquad G(v) = \int_{-1}^{v} g(\theta) \, \mathrm{d}\theta.$$

Then g is continuous and bounded on $[-1, 1]$, $G(-1) = 0$, $G(1) = m_s$, and

$$g(v) \leq C(1 - |v|)^{1+1/\beta}, \tag{6.3}$$

$$G(v) \leq C(1 + v)^{2+1/\beta} \qquad (v \leq 0),$$

$$m_s - G(v) \leq C(1 - v)^{2+1/\beta} \qquad (v \geq 0).$$

At all sufficiently large heights,

$$N(z) := \int_0^\infty G(U(r, z)) \, \mathrm{d}r = m_s R(z) + O(1), \tag{6.4}$$

and

$$e(z) := \sup_{r \geq 0} | - U_r(r, z) - g(U(r, z))| \longrightarrow 0. \tag{6.5}$$

Proof. Lemma 2.1 gives $0 < m_s < \infty$ and the two-sided phase tail for every $0 < s < 1$. Its lower bound $1 - |q(t)| \geq c_0|t|^{-\beta}$ and $q'(t) \leq C|t|^{-1-\beta}$ imply (6.3) near each endpoint. Compactness gives the bound on the remaining interval, and integration gives the two primitive estimates. The decay of q' gives continuity of g at the endpoints; the change of variable $v = q(t)$ gives $G(1) = m_s$.

By (4.8) and the primitive estimates,

$$|G(U(r,z)) - m_s \mathbf{1}_{\{r<R(z)\}}| \leq C(1 + |r - R(z)|)^{-1-2\beta}.$$

This bound is uniformly integrable for every $\beta > 0$, proving (6.4). Further, (6.3) and (4.8) give $g(U) \leq C(1 + |r - R|)^{-1-\beta}$, the same bound as for $|U_r|$. On each fixed strip $|r - R| \leq M$, the local C^1 convergence in (4.7) and $g(q(-\xi)) = q'(-\xi)$ give uniform matching. Outside the strip both terms are bounded by $C(1 + M)^{-1-\beta}$. First let $z \to \infty$, then $M \to \infty$. □

Lemma 6.4 (Integrated drift)**.** *For every fixed sufficiently large Z, put*

$$B(z) = \int_0^\infty U_z(r,z)(-U_r(r,z))\,\mathrm{d}r.$$

Then

$$\int_Z^z B(t)\,\mathrm{d}t = m_s R(z) + o(R(z)). \tag{6.6}$$

Proof. The integral defining B is finite and locally bounded, since U_z is bounded and $\int_0^\infty (-U_r)\,\mathrm{d}r \leq 2$. The chain rule, nonnegativity and Tonelli's theorem give

$$N(z_2) - N(z_1) = \int_{z_1}^{z_2} \int_0^\infty g(U)U_z\,\mathrm{d}r\,\mathrm{d}t.$$

Both sides are finite by Lemma 6.3; this also proves local absolute continuity of N.

Fix $\varepsilon > 0$ and choose $Z_\varepsilon \geq Z$ so that $e(t) \leq \varepsilon$ for $t \geq Z_\varepsilon$. At a final height $z \geq Z_\varepsilon$, set $A = 2R(z)$ and hold this cutoff fixed during the height integration. On $0 \leq r \leq A$,

$$\int_{Z_\varepsilon}^z \int_0^A U_z| - U_r - g(U)|\,\mathrm{d}r\,\mathrm{d}t \leq \varepsilon \int_0^A [U(r,z) - U(r,Z_\varepsilon)]\,\mathrm{d}r \leq 4\varepsilon R(z).$$

For $r > A$ and $Z_\varepsilon \leq t \leq z$, the increasing radius gives $r - R(t) > r/2$. Equations (6.2) and (6.3), together with (4.8), bound the difference by $Cr^{-1-\beta}$. Since $\int_{Z_\varepsilon}^z U_z(r,t)\,\mathrm{d}t \leq 2$,

$$\int_{Z_\varepsilon}^z \int_A^\infty U_z| - U_r - g(U)|\,\mathrm{d}r\,\mathrm{d}t \leq C \int_A^\infty r^{-1-\beta}\,\mathrm{d}r \leq CR(z)^{-\beta}.$$

It follows that

$$\left| \int_{Z_\varepsilon}^z B(t)\,\mathrm{d}t - N(z) + N(Z_\varepsilon) \right| \leq 4\varepsilon R(z) + CR(z)^{-\beta}.$$

The earlier fixed height interval contributes a finite constant. Divide by $R(z)$, use (6.4), and let $\varepsilon \downarrow 0$. □

6.3. **The energy at the axis.** Let $V = P_y^{(2)} * U$ and define

$$E_0(z) = \frac{d_s}{2} \int_0^\infty y^{1-\beta}(V_z^2 + V_y^2)(0,z,y)\,\mathrm{d}y + F(U(0,z)). \tag{6.7}$$

Its leading coefficient depends only on the extension of a unit interval. We use Euler's beta function $\mathrm{B}(u,v) = \Gamma(u)\Gamma(v)/\Gamma(u+v)$ in the calculation.

Lemma 6.5 (Sharp axis energy)**.** *For every $0 < s < 1$,*

$$R(z)^{2s} E_0(z) \longrightarrow \kappa_s = \frac{2\Gamma(s + 1/2)}{\sqrt{\pi}\Gamma(1 - s)}. \tag{6.8}$$

Proof. Write $R_z = R(z)$. Since $R' = o(1)$ and $R = o(z)$,

$$R(z + R_z T)/R_z \longrightarrow 1$$

uniformly for bounded T. Indeed, the interval $[z-CR_z, z+CR_z]$ tends to infinity, and the mean value estimate uses the supremum of R' on that interval. For fixed (X,T) with $|X| \neq 1$, this implies, eventually,

$$|R_z|X| - R(z+R_zT)| \geq \tfrac{1}{2}R_z||X|-1|.$$

The sign of $R(z+R_zT) - R_z|X|$ is also eventually that of $1-|X|$. Thus (4.8) gives, almost everywhere on $\mathbb{R}^2$,

$$u_z(X,T) := U(R_zX, z+R_zT) \longrightarrow u_*(X,T) := 2\mathbf{1}_{\{|X|<1\}} - 1.$$

The rescaled extension $W_z(X,T,\eta) = V(R_zX, z+R_zT, R_z\eta)$ is the extension of u_z. The energy scales exactly as

$$R_z^\beta \int_0^\infty y^{1-\beta}(V_z^2+V_y^2)(0,z,y)\,\mathrm{d}y = \int_0^\infty \eta^{1-\beta}((W_z)_T^2 + (W_z)_\eta^2)(0,0,\eta)\,\mathrm{d}\eta.$$

At every $\eta > 0$ the derivative kernels are integrable, so bounded convergence gives convergence of its first derivatives at $(0,0,\eta)$ to those of W_*, including $(W_z)_T \to 0$. The bound C/η makes the weighted squared derivatives converge on each compact subinterval of $(0,\infty)$.

We verify tightness at both endpoints. Fix $0 < \rho < 1/16$. For large z, the relevant height interval lies above the threshold where $|R'| \leq 1$. Thus, on $B_{4\rho R_z}(0,z)$,

$$R(z') - |x'| \geq (1-8\rho)R_z \geq R_z/2.$$

Equation (4.8) therefore gives

$$\|1-U\|_{C^2(B_{4\rho R_z}(0,z))} \leq CR_z^{-\beta}.$$

Choose a smooth cutoff $0 \leq \chi \leq 1$ equal to one on $B_{2\rho R_z}(0,z)$ and zero outside $B_{3\rho R_z}(0,z)$, with the usual scaled derivative bounds. Split $1-U = v_0 + v_1$, $v_0 = \chi(1-U)$, and let $V_i = P_y^{(2)} * v_i$, so that $V_0 + V_1 = 1 - V$. Then $\|v_0\|_{C^2(\mathbb{R}^2)} \leq \delta_z = CR_z^{-\beta}$. Apply Lemma 5.1 at arbitrary D with $K = \delta = \delta_z$, then let $D \to \infty$. It gives

$$\int_0^\infty y^{1-\beta}|\nabla V_0(0,z,y)|^2\,\mathrm{d}y \leq C\delta_z^2.$$

Since $R_z^\beta \delta_z^2 \to 0$, this near part has vanishing scaled energy.

The remote part v_1 is bounded and vanishes in $B_{2\rho R_z}(0,z)$. For $0 < y \leq \rho R_z$, direct differentiation of the Poisson kernel gives at the axis

$$|(V_1)_y| \leq Cy^{\beta-1}R_z^{-\beta}, \qquad |(V_1)_z| \leq Cy^\beta R_z^{-1-\beta}.$$

These follow by integrating the kernel bounds $Cy^{\beta-1}|X|^{-2-\beta}$ and $Cy^\beta|X|^{-3-\beta}$ outside that ball. Therefore, for $0 < \varepsilon < \rho$,

$$\limsup_{z\to\infty} R_z^\beta \int_0^{\varepsilon R_z} y^{1-\beta}(V_y^2+V_z^2)(0,z,y)\,\mathrm{d}y \leq C(\varepsilon^\beta + \varepsilon^{2+\beta}). \tag{6.9}$$

The near part was estimated in its full energy norm, so its scaled energy tends to zero; its mixed term with the remote part on this interval also tends to zero by Cauchy–Schwarz. At the other endpoint, $|\nabla V| \leq C/y$ gives

$$R_z^\beta \int_{MR_z}^\infty y^{1-\beta}(V_y^2+V_z^2)(0,z,y)\,\mathrm{d}y \leq CM^{-\beta}. \tag{6.10}$$

The two estimates prove tightness. Also, $1 - U(0,z) = O(R_z^{-\beta})$ and $F(1) = F'(1) = 0$ imply $R_z^\beta F(U(0,z)) \to 0$.

The limiting trace is independent of T. Dimension reduction of the normalized Poisson kernel gives, with $p_s = \Gamma(s+1/2)/(\sqrt{\pi}\Gamma(s))$,

$$W_*(0,\eta) = 4p_s\int_0^{1/\eta}(1+t^2)^{-s-1/2}\,\mathrm{d}t - 1, \qquad (W_*)_\eta(0,\eta) = -4p_s\frac{\eta^{2s-1}}{(1+\eta^2)^{s+1/2}}.$$

Consequently the limit in (6.8) equals

$$\frac{d_s}{2}\int_0^\infty \eta^{1-2s}(W_*)_\eta^2\,\mathrm{d}\eta = 4d_s p_s^2 \mathrm{B}(s,s+1) = \frac{2\Gamma(s+1/2)}{\sqrt{\pi}\Gamma(1-s)}.$$

The last equality uses $\Gamma(s+1)=s\Gamma(s)$ and the duplication formula. □

6.4. **Renormalized stress and the height law.** For $s \le 1/2$, the flux integrated over the full radial half-line need not converge absolutely at large extension heights. Its difference between two physical heights has an additional power of decay.

Lemma 6.6 (Renormalized planar stress)**.** *For a fixed height Z, put*

$$K_y(z) = \int_0^\infty V_r(r,z,y)V_z(r,z,y)\,\mathrm{d}r, \qquad \mathcal{J}(z) = d_s\int_0^\infty y^{1-\beta}[K_y(z)-K_y(Z)]\,\mathrm{d}y.$$

The latter integral converges and is continuous, and

$$c\int_Z^z B(t)\,\mathrm{d}t = \int_Z^z E_0(t)\,\mathrm{d}t + \mathcal{J}(z). \tag{6.11}$$

For $h = z - Z \ge 0$,

$$|\mathcal{J}(z)| \le C\begin{cases} 1+h^{1-\beta}, & 0<\beta<1,\\ 1+\log(1+h), & \beta=1,\\ 1, & 1<\beta<2. \end{cases} \tag{6.12}$$

Proof. Poisson convolution preserves evenness. The radial order and lateral phase arguments at the start of the proof of Lemma 5.2 also apply when $n=1$ and $0<s<1$, giving $V_r \le 0$ and

$$\int_0^\infty (-V_r)\,\mathrm{d}r = V(0,z,y)+1 \le 2.$$

For $y \ge 1$, differentiated Poisson-kernel norms give

$$\|V_z\|_\infty \le C/y, \qquad \|V_{zz}\|_\infty \le C/y^2, \qquad \int_{\mathbb{R}} |V_{xz}|\,\mathrm{d}x \le C/y.$$

For the last bound write $V_{xz} = \partial_z P_y^{(2)} * U_x$, integrate absolutely first in x, and use $\int_{\mathbb{R}} |U_x(x,t)|\,\mathrm{d}x \le 4$ at each trace height, together with $\|\partial_z P_y^{(2)}\|_1 \le C/y$. Hence

$$|K_y| \le C/y, \qquad |K_y'| \le C/y^2 \quad (y \ge 1,\ \text{a.e. in } z).$$

The two differentiated product terms have absolute radial integrals bounded by

$$\|V_z\|_\infty \int_0^\infty |V_{rz}|\,\mathrm{d}r \le C/y^2, \qquad \|V_{zz}\|_\infty \int_0^\infty |V_r|\,\mathrm{d}r \le C/y^2.$$

Apply the fundamental theorem of calculus first on $0<r<L$. On every compact physical height interval these bounds give integrability in (r,z); Fubini's theorem and then $L\to\infty$ show that K_y is locally absolutely continuous and satisfies the displayed derivative bound almost everywhere. For $y \le 1$, bounded V_z and radial variation give $|K_y| \le C$. Consequently

$$|K_y(z)-K_y(Z)| \le C\min(y^{-1}, |z-Z|y^{-2}) \quad (y \ge 1). \tag{6.13}$$

The small-height bound $|K_y| \le C$ is integrable with weight $y^{1-\beta}$. At large heights, for $h = |z-Z|$,

$$\int_1^\infty y^{1-\beta}|K_y(z)-K_y(Z)|\,\mathrm{d}y \le C\int_1^\infty \min\{y^{-\beta}, hy^{-1-\beta}\}\,\mathrm{d}y.$$

For $h \ge 1$, split this integral at $y=h$; for $h<1$, use $hy^{-1-\beta}$. This proves convergence and (6.12), with locally uniform tails in z. For fixed y, continuity follows from local smooth convergence and the radial tail bound $C\min(1,y^{-1})[V(L,z,y)+1] \to 0$, locally uniformly in z.

We derive (6.11) before removing the extension cutoffs. Set

$$T_{\varepsilon,Y}(r,z) = \frac{d_s}{2}\int_\varepsilon^Y y^{1-\beta}(V_r^2 - V_z^2 - V_y^2)\,\mathrm{d}y - F(U(r,z)),$$
$$P_{\varepsilon,Y}(r,z) = d_s\int_\varepsilon^Y y^{1-\beta}V_rV_z\,\mathrm{d}y.$$

The extension equation gives the exact cutoff stress identity

$$\partial_r T_{\varepsilon,Y} + \partial_z P_{\varepsilon,Y} = -d_s[\,y^{1-\beta}V_yV_r\,]_\varepsilon^Y - f(U)U_r.$$

Integrate over $r \in (0,L)$ and physical height $t \in (Z,z)$, then let $L \to \infty$ with ε, Y fixed. The lateral stress vanishes by the lateral phase and the Poisson formula. At the axis $V_r = 0$, so $E_{\varepsilon,Y}(t) := -T_{\varepsilon,Y}(0,t)$ is the cutoff axis energy, and the integrated identity is

$$\int_Z^z E_{\varepsilon,Y}(t)\,\mathrm{d}t + d_s\int_\varepsilon^Y y^{1-\beta}[K_y(z) - K_y(Z)]\,\mathrm{d}y$$
$$= -d_s\int_Z^z\left[y^{1-\beta}\int_0^\infty V_yV_r\,\mathrm{d}r\right]_\varepsilon^Y\mathrm{d}t - \int_Z^z\int_0^\infty f(U)U_r\,\mathrm{d}r\,\mathrm{d}t.$$

The upper extension boundary product, integrated in r, is at most

$$CY^{1-\beta}(C/Y)\int_0^\infty |V_r|\,\mathrm{d}r \le CY^{-\beta},$$

uniformly on the fixed physical height interval.

At the lower boundary, the conormal formula gives $d_s\varepsilon^{1-\beta}V_y \to f(U) - cU_z$ uniformly, while $V_r \to U_r$ in $L^1(0,\infty)$ for each fixed t by the radial-mass argument in the proof of Lemma 5.2, with $a_0 = 0$. That argument uses only local convergence, nonnegativity of $-V_r$ and convergence of its total mass, so it applies at every order. The mass bound 2 permits dominated convergence in t. The lower boundary integral therefore tends to $\int(f(U) - cU_z)U_r\,\mathrm{d}r$. This cancels the potential derivative and leaves cB. At the axis, (2.7) gives the integrable small-height majorant $C(y^{1-\beta} + y^{\beta-1})$ and large-height majorant $Cy^{-1-\beta}$ for the energy density, uniformly on the physical height interval. Letting $\varepsilon \downarrow 0$ and $Y \to \infty$ yields the difference of the two fluxes, which converges by (6.13). This is precisely (6.11). □

Proof of Theorem 1.5. All the preceding lemmas apply to every wave in the stated class. Set

$$S(z) = \int_Z^z R(t)^{-\beta}\,\mathrm{d}t.$$

We first show $S \to \infty$ and $\mathcal{J} = o(S)$. If $0 < \beta < 1$, the increasing radius gives $S(z) \ge (z - Z)R(z)^{-\beta}$, whence

$$\frac{z^{1-\beta}}{S(z)} \le C\left(\frac{R(z)}{z}\right)^\beta \longrightarrow 0.$$

If $\beta = 1$, for each $\varepsilon > 0$ one has $R(t) \le \varepsilon t$ at all late heights. Integrating $R(t)^{-1} \ge \varepsilon^{-1}t^{-1}$ proves $S(z)/\log z \to \infty$. Thus (6.12) suffices in both low-order cases. If $\beta > 1$, the flux is bounded. Equations (6.11) and (6.6) force $\int_Z^z E_0\,\mathrm{d}t \to \infty$, since $cm_sR \to \infty$. Equation (6.8) gives $E_0(t) \le 2\kappa_sR(t)^{-\beta}$ at large heights, so $S \to \infty$ and again $\mathcal{J} = o(S)$.

Since $S \to \infty$, integrating $E_0 = (\kappa_s + o(1))R^{-\beta}$ gives $\int_Z^z E_0 = \kappa_sS + o(S)$: split the error integral at a fixed large height as in Section 5. The integrated identity now becomes

$$cm_sR + o(R) = \kappa_sS + o(S).$$

Equivalently, $(cm_s + o(1))R = (\kappa_s + o(1))S$; both leading coefficients are positive, so

$$\frac{R(z)}{S(z)} \longrightarrow A := \frac{\kappa_s}{cm_s}.$$

Lemma 5.4 with $\gamma = \beta$ yields

$$\frac{R(z)^{1+\beta}}{z} \longrightarrow (1+\beta)A. \tag{6.14}$$

For every fixed $\theta \in (-1,1)$, (4.11) gives $R_\theta/R \to 1$. Together with (6.14),

$$\frac{R_\theta(z)^{1+\beta}}{z} \longrightarrow (1+\beta)A.$$

Using $H_\theta(r) \to \infty$ and (4.10), substitution into the last radius limit gives

$$\frac{H_\theta(r)}{r^{1+2s}} \longrightarrow \frac{cm_s}{(1+2s)\kappa_s}.$$

Finally $\kappa_{1/2} = 2/\pi$, so the critical coefficient is $\pi c m_{1/2}/4$. □

7. Higher dimensional interfaces at low orders

Throughout this section $n \geq 2$, $0 < s \leq 1/2$, $c > 0$, and U is any bounded classical solution of (1.2) with the orders and locally uniform phase limits in (1.3)–(1.4). We assume (A), keep the stable levels in (2.3), and normalize $q(0) = \alpha$. Write $d = n+1$, $\beta = 2s$, $R = R_\alpha$, and $h(z) = R(z)/2$ whenever this radius is defined. The constants m_s, d_s and the extension V have the normalizations fixed above.

The planar Liouville theorem also describes translates of a cylindrical wave far from its axis. For $s < 1/2$, the resulting force is the fractional curvature of the transverse unit ball. At $s = 1/2$, its leading term comes from extension heights between the transition width and the radius.

7.1. **Preparation and integrated drift.**

Lemma 7.1 (Preparation in higher dimensions)**.** *Under the assumptions of this section, $R \to \infty$, $R' > 0$, $R' \to 0$, $R = o(z)$, and*

$$U(R(z)e_1 + X, z+T) \longrightarrow q(-X_1) \quad \text{locally in } C^2(\mathbb{R}^{n+1}).$$

For all large z and $r \geq 0$,

$$|U(r,z) - \operatorname{sgn}(R(z)-r)| + |D^2U(r,z)| \leq C(1+|r-R(z)|)^{-\beta},$$
$$|\nabla U(r,z)| \leq C(1+|r-R(z)|)^{-1-\beta}.$$

Derivatives are Cartesian, including at the axis. For every fixed $\theta \in (-1,1)$, the unique positive radius $R_\theta(z)$ satisfying $U(R_\theta(z),z) = \theta$ is defined at all sufficiently large heights, and $R = R_\alpha$. It obeys

$$R_\theta(z) - R(z) \longrightarrow -q^{-1}(\theta).$$

Proof. All assertions except the sharper first-derivative bound are supplied by Lemma 4.4, with exactly the present potential and wave hypotheses. For that bound, apply Lemma 6.1 in dimension $n+1$, using its coordinate variations at most 4 transversely and 2 vertically, and the core-distance bound (4.9). The global derivative bound handles a fixed-width neighborhood of the interface. Horizontal rotation includes U_r, and the Cartesian formulation covers the axis. □

Lemma 7.2 (Integrated drift with a moving cut)**.** *Under the assumptions of this section put $h(z) = R(z)/2$ and*

$$B_h(z) = \int_{h(z)}^{\infty} U_z(-U_r)\,\mathrm{d}r.$$

For every sufficiently large fixed Z,

$$\int_Z^z B_h(t)\,\mathrm{d}t = m_s R(z) + o(R(z)).$$

Proof. The proofs of Lemmas 6.3 and 6.4 use only the layer tails, local layer convergence, strict radial and vertical order, and the phase and radial derivative bounds on each half-line. They use no dimension-specific equation after those inputs are established. Lemma 7.1 supplies all of them here, with the same q and $\beta = 2s$. Thus their primitive and fixed-cutoff matching argument gives, for $B(t) = \int_0^\infty U_z(r,t)(-U_r(r,t))\,\mathrm{d}r$,

$$\int_Z^z B(t)\,\mathrm{d}t = m_s R(z) + o(R(z)).$$

Finally on $r \le R(t)/2$, $R(t) - r \ge r$, so the omitted part satisfies

$$0 \le \int_Z^z (B - B_h)\,\mathrm{d}t \le C \int_0^\infty (1+r)^{-1-\beta} \int_Z^z U_z(r,t)\,\mathrm{d}t\,\mathrm{d}r \le C.$$

No unweighted phase mass is used. □

7.2. **The renormalized stress balance.**

Lemma 7.3 (Renormalized stress outside a moving radius)**.** *Fix a sufficiently large Z and, for $z \ge Z$, set*

$$h(z) = R(z)/2, \qquad B_h(z) = \int_{h(z)}^\infty U_z(-U_r)\,\mathrm{d}r.$$

Then $h' = R'/2$ is bounded and tends to zero. Define

$$A(r,z) = d_s \int_0^\infty y^a V_r^2\,\mathrm{d}y, \quad P(r,z) = d_s \int_0^\infty y^a V_r V_z\,\mathrm{d}y,$$

$$T(r,z) = \frac{d_s}{2} \int_0^\infty y^a (V_r^2 - V_z^2 - V_y^2)\,\mathrm{d}y - F(U(r,z)), \quad I_h(z) = \int_{h(z)}^\infty A(r,z)/r\,\mathrm{d}r.$$

These quantities are finite. With

$$K_y(z) = \int_{h(z)}^\infty V_r V_z\,\mathrm{d}r, \qquad \mathcal{J}(z) = d_s \int_0^\infty y^a [K_y(z) - K_y(Z)]\,\mathrm{d}y,$$

the subtracted flux converges and

$$c \int_Z^z B_h(t)\,\mathrm{d}t = \int_Z^z \{(n-1)I_h - T(h(t),t) + h'(t)P(h(t),t)\}\,\mathrm{d}t + \mathcal{J}(z).$$

Moreover

$$|T(h,z)| + |P(h,z)| \le C R(z)^{-\beta},$$

$$|\mathcal{J}(z)| \le C \begin{cases} 1 + z^{1-\beta}, & 0 < \beta < 1, \\ 1 + \log(1+z), & \beta = 1. \end{cases}$$

The extension is radially decreasing and vertically increasing, with the inherited lower lateral phase; more precisely, for every $y > 0$,

$$V_r \le 0, \quad V_z \ge 0, \quad \lim_{r\to\infty} V(r,z,y) = -1, \qquad \int_0^\infty (-V_r)\,\mathrm{d}r \le 2,$$

$$\int_0^\infty V_r^2\,\mathrm{d}r \le C \min(1, y^{-1}), \qquad \int_{h(z)}^\infty V_r^2/r\,\mathrm{d}r \le C/y^2.$$

For all $y > 0$, $|\nabla V| \le C/y$ and $|D_{x,z}^2 V| \le C/y^2$; for $0 < y \le 1$ one also has $|\nabla_{x,z} V| \le C$ and $|V_y| \le C y^{\beta-1}$.

Proof. We use Lemma 7.1. The Poisson order and lateral phase arguments in the proof of Lemma 5.2 apply in every dimension and for every $0 < s < 1$. Hence $V_z \ge 0$, $V_r \le 0$, $V(r,z,y) \to -1$ as $r \to \infty$, and

$$\int_0^\infty (-V_r(r,z,y))\,\mathrm{d}r = V(0,z,y) + 1 \le 2.$$

Normalized Poisson derivative norms give $|\nabla V| \le C/y$, $|D_{x,z}^2 V| \le C/y^2$ for every $y > 0$; for $y \le 1$, tangential derivatives are bounded and $|V_y| \le C y^{\beta-1}$ by (2.7). The tangential derivative bound and radial variation imply $\int_0^\infty V_r^2\,\mathrm{d}r \le C \min(1, y^{-1})$. Thus A, P, T converge at each physical point. The cut satisfies $h' = R'/2 = o(1)$ by Lemma 7.1, and B_h is finite and locally bounded because U_z is bounded and $\int(-U_r) \le 2$.

For later use there is the stronger radial energy bound, valid for every $y > 0$,

$$\int_h^\infty \frac{V_r^2}{r}\,\mathrm{d}r \le C/y^2.$$

Indeed radial symmetry gives $V_r(0) = 0$ and $|V_r|/r \le C/y^2$ for $r \le y$; for $r \ge y$ use $1/r \le 1/y$ and $|V_r| \le C/y$. In each part multiply the resulting bound by $\int |V_r| \le 2$. For $y \le 1$ one also has

$\int_h^\infty V_r^2/r\,\mathrm{d}r \le C/h$ by trace derivative bounds and variation. These estimates prove finiteness of I_h.

Every x_1-line has trace variation at most 4, uniformly over all other coordinates, by Lemma 6.1. Write $H = (H_1, H_\perp, H_z)$ and $V_{x_1 z} = \partial_z P_y^{(d)} * U_{x_1}$. Tonelli and this uniform bound give

$$\begin{aligned}\int_{\mathbb{R}} |V_{x_1 z}(r,0,z,y)|\,\mathrm{d}r &\le \int_{\mathbb{R}^d} |\partial_z P_y^{(d)}(H)| \int_{\mathbb{R}} |U_{x_1}(r-H_1, -H_\perp, z-H_z)|\,\mathrm{d}r\,\mathrm{d}H \\ &\le 4\|\partial_z P_y^{(d)}\|_1 \le C/y.\end{aligned}$$

Together with $\|V_z\|_\infty \le C/y$ and $\|V_{zz}\|_\infty \le C/y^2$, differentiation of the moving radial integral gives, for $y \ge 1$,

$$|K_y| \le C/y, \qquad |K_y'| \le C/y^2 + C|h'|/y^2 \le C/y^2 \quad \text{a.e. in } z.$$

The moving-boundary term is $-h'V_r(h)V_z(h)$, bounded by $C\|h'\|_\infty/y^2$. To justify the derivative formula, first use Leibniz' rule on $h(z) < r < L$. The two interior differentiated terms have absolute radial integrals bounded by

$$\|V_z\|_\infty \int_0^\infty |V_{rz}|\,\mathrm{d}r, \qquad \|V_{zz}\|_\infty \int_0^\infty |V_r|\,\mathrm{d}r,$$

each at most C/y^2, uniformly in z. Fubini's theorem on a compact physical height interval permits $L \to \infty$ in the integrated derivative formula. Hence K_y is locally absolutely continuous, with the asserted bound for K_y' almost everywhere. In particular, for $\ell = z - Z \ge 0$,

$$|K_y(z) - K_y(Z)| \le C\min(y^{-1}, \ell y^{-2}) \quad (y \ge 1),$$

whereas for $y \le 1$ each K_y is bounded. Splitting at $y = \max(1, \ell)$ proves flux convergence and the stated growth bounds.

The extension equation is

$$V_{rr} + (n-1)r^{-1}V_r + V_{zz} + y^{-a}(y^a V_y)_y = 0, \quad -d_s \lim_{y\downarrow 0} y^a V_y = cU_z - f(U).$$

Multiplying it by $d_s y^a V_r$ and integrating in y yields the distributional identity $\partial_r T + \partial_z P = -(n-1)A/r - cU_z U_r$. To justify its integrated use, first restrict to $h(t) < r < L$, $Z < t < z$, $\varepsilon < y < Y$. At fixed extension cutoffs, let $L \to \infty$. The lateral stress tends to zero by the lateral phase and Poisson formula; radial tails of each cross flux are bounded by $C\min(1, y^{-1})[V(L,t,y)+1]$ uniformly on compact time intervals. The upper extension boundary product is at most $CY^{1-\beta}(1/Y)\int |V_r| \le CY^{-\beta}$ per time slice.

The global C^2 bound and the uniform conormal convergence proved after (2.7) give $-d_s\varepsilon^{1-\beta}V_y \to L_{d,s}U = cU_z - f(U)$ uniformly in the physical space. For each fixed t, the radial-mass argument in the proof of Lemma 5.2, with $a_0 = h(t)$, gives $V_r(\cdot,t,\varepsilon) \to U_r(\cdot,t)$ in $L^1(h(t),\infty)$. It uses only local convergence, nonnegativity and convergence of the total masses, so applies at every order. The uniform mass bound 2 and the conormal bound permit dominated convergence in t for their product. Consequently the potential term cancels and leaves cB_h. The moving boundary flux is $-T(h) + h'P(h)$. The preceding bounds justify $\varepsilon \downarrow 0$ and $Y \to \infty$ and produce the subtracted flux, not either unsubtracted improper flux separately.

Finally a ball of radius $\rho R(z)$ about $(h(z)e_1, z)$ remains in the upper phase, at distance comparable to $R(z)$ from the transition set, for fixed sufficiently small ρ. This follows from $R' \to 0$ and $R = o(z)$. The preparation bounds give $\|1-U\|_{C^2}$ on that ball at most $CR^{-\beta}$. Apply Lemma 5.1, valid in every dimension and at every order, with $D = \rho R/4 \ge 1$, $K = 2$ and $\delta = CR^{-\beta} \le 2$ at large heights. Its slice energy is $O(R^{-2\beta} + R^{-\beta}) = O(R^{-\beta})$. Cauchy–Schwarz bounds $T(h)$ and $P(h)$; the remaining potential term is $F(U(h)) = O(R^{-2\beta})$ by its quadratic zero. □

7.3. **The transverse ball and the strongly nonlocal force.**

Lemma 7.4 (The force of a transverse unit ball)**.** *For $n \ge 2$ and $0 < s < 1/2$, let $w = 2\mathbf{1}_{B_1} - 1$ on $\mathbb{R}^n$, and let $W = P_y^{(n)} * w$. Put*

$$A_*(r) = d_s \int_0^\infty y^a W_r^2 \,\mathrm{d}y, \qquad T_*(r) = \frac{d_s}{2} \int_0^\infty y^a (W_r^2 - W_y^2) \,\mathrm{d}y.$$

Here A_* is the nonnegative measurable slice energy, allowed to be infinite at $r = 1$; T_* is defined for $r \ne 1$. For every $0 < h < 1$,

$$(n-1) \int_h^\infty \frac{A_*(r)}{r} \,\mathrm{d}r - T_*(h) = \kappa_{n,s},$$
$$\kappa_{n,s} = \frac{2\Gamma(n/2+s)\Gamma(1/2-s)}{\sqrt{\pi}\Gamma(1-s)\Gamma(n/2-s)}.$$

Proof. We keep the normalization in (2.1) and (2.6), and put $B = B_1$. For $\beta = 2s < 1$, the fractional perimeter $\mathrm{Per}_\beta(B) = \int_B \int_{B^c} |x-t|^{-n-\beta} \,\mathrm{d}t \,\mathrm{d}x$ is finite. Indeed, changing variables to the displacement H gives

$$2\,\mathrm{Per}_\beta(B) = \int_{\mathbb{R}^n} \frac{|B \triangle (B-H)|}{|H|^{n+\beta}} \,\mathrm{d}H.$$

For $|H| < 1$, the symmetric difference lies in the annulus $1 - |H| < |x| < 1 + |H|$, whose volume is at most $C_n|H|$. For $|H| \ge 1$, its volume is at most $2|B|$. Consequently

$$\mathrm{Per}_\beta(B) \le C_n \left\{ \int_0^1 \rho^{-\beta} \,\mathrm{d}\rho + \int_1^\infty \rho^{-1-\beta} \,\mathrm{d}\rho \right\} < \infty.$$

To justify smooth approximation in precisely this energy, put

$$\mathcal{F}(x,H) = \frac{w(x+H) - w(x)}{|H|^{(n+\beta)/2}}.$$

The finite cross-kernel integral says $\mathcal{F} \in L^2(\,\mathrm{d}x\,\mathrm{d}H)$. For a nonnegative smooth compact mollifier ρ_δ of mass one, the difference quotient of $w_\delta = \rho_\delta * w$ is $\rho_\delta *_x \mathcal{F}$. Translation continuity in this L^2 space, followed by Jensen or the triangle inequality, gives $\|\rho_\delta *_x \mathcal{F} - \mathcal{F}\|_2 \to 0$. Thus $w_\delta \to w$ in the homogeneous H^s energy. The smooth extension energy identity passes to the completion. Also bounded almost everywhere convergence of w_δ implies that its Poisson extensions and their gradients converge at positive heights to W; consequently that completion is exactly the stated convolution. The normalized extension energy is therefore finite and equals

$$\mathcal{E}(W) = \frac{d_s}{2} \int_{\mathbb{R}^{n+1}_+} y^a |\nabla W|^2 \,\mathrm{d}x \,\mathrm{d}y = \frac{C_{n,s}}{4} \iint \frac{|w(x) - w(t)|^2}{|x-t|^{n+\beta}} \,\mathrm{d}x \,\mathrm{d}t = 2C_{n,s}\,\mathrm{Per}_\beta(B).$$

For a compact interval $I \subset [0,1)$, the trace is constant in a uniform neighborhood of each $r \in I$. Direct differentiation of the remote Poisson kernel gives $|W_r| \le Cy^\beta$ and $|W_y| \le Cy^{\beta-1}$ for $0 < y \le 1$, uniformly on I; for $y \ge 1$ all first derivatives are at most C/y. Thus the weighted square majorants are $C(y^{1+\beta} + y^{\beta-1})$ near zero and $Cy^{-1-\beta}$ at infinity. They are integrable, so $T_*(h)$ is finite and continuous for $0 \le h < 1$ by dominated convergence. The exterior is handled just as explicitly. Fix $1 < r_0 \le r \le r_1 < \infty$ and put $\delta = r_0 - 1 > 0$. Since $W + 1 = P_y^{(n)} * (2\mathbf{1}_{B_1})$, its trace is constant -1 near these evaluation points and the nonconstant source stays at distance at least δ. Differentiated remote kernel bounds give $|\nabla_x W| \le Cy^\beta$ and $|W_y| \le Cy^{\beta-1}$ for $0 < y < \min(1, \delta/2)$, uniformly for $r \in [r_0, r_1]$. The weighted square majorants are again $C(y^{1+\beta} + y^{\beta-1})$ at zero. On any remaining compact positive height interval the derivatives are uniformly bounded, while $y \ge 1$ has the integrable majorant $Cy^{-1-\beta}$. Dominated convergence therefore proves finiteness and continuity of $T_*(r)$ also on every compact exterior interval, as required for all $r \ne 1$.

We justify the zero-trace completion used for the variational argument. If v has finite weighted gradient norm, is smooth for $y > 0$, and has zero trace almost everywhere, the one-dimensional weighted Hardy estimate gives

$$\int_{\mathbb{R}^n} \int_0^\infty y^{a-2} |v|^2 \,\mathrm{d}y \,\mathrm{d}x \le \frac{4}{(1-a)^2} \int_{\mathbb{R}^n} \int_0^\infty y^a |v_y|^2 \,\mathrm{d}y \,\mathrm{d}x.$$

Here is its direct proof in the present class. For almost every x, vertical Cauchy–Schwarz and the zero boundary limit give

$$y^{a-1}|v(x,y)|^2 \le \frac{1}{1-a}\int_0^y t^a|v_y(x,t)|^2\,\mathrm{d}t \longrightarrow 0 \quad (y\downarrow 0).$$

Integrate the derivative of $y^{a-1}v^2$ on a finite vertical interval, let its lower endpoint tend to zero, discard its nonpositive upper boundary contribution after dividing by $a-1<0$, and apply Cauchy–Schwarz to $\int y^{a-1}vv_y$. This gives the displayed Hardy constant; truncation and monotone convergence justify the argument if the left side is not initially known finite. Integration in x gives the full estimate. First cutting v off on $y<\varepsilon$, with transition in $(\varepsilon,2\varepsilon)$, changes its gradient norm by a quantity tending to zero: the extra derivative is controlled by the Hardy mass on that strip. Next cut off at full Euclidean radius L, with transition in the annulus $L<|(x,y)|<2L$. Since $y\le 2L$ there, the extra squared derivative is bounded by $Cy^{a-2}|v|^2$, whose tail tends to zero. The gradient tails themselves also vanish by finite energy. These two cutoffs leave a compactly supported function away from $y=0$, where the weight is smooth and bounded above and below; ordinary smoothing proves density of interior compact smooth tests in this zero-trace energy class.

The extension equation gives zero mixed energy against each such test; density and Cauchy–Schwarz extend this orthogonality to every zero-trace finite energy variation. Expanding the square proves that W minimizes weighted energy in its affine trace class. The same argument applies to the Poisson extension of every smooth diffeomorphic deformation of w: its trace has finite homogeneous energy by the bi-Lipschitz change of variables, its Poisson extension has that energy by completion, and the differences used below have zero trace almost everywhere.

Take a smooth bounded radial vector field $X(x)=\eta(r)r^{1-n}e_r$, supported away from the origin and initially also cut off at infinity, with $\eta=1$ near $r=1$. Let $\Phi_\varepsilon=\mathrm{Id}+\varepsilon X$ and deform the trace to $w\circ\Phi_\varepsilon^{-1}$. The derivative of its minimum extension energy is the derivative of the energy of $W(\Phi_\varepsilon^{-1}(x),y)$ at zero. Here is a justification of this envelope assertion. Pull every deformed minimizer back by Φ_ε. It has the fixed trace w and minimizes a weighted coercive quadratic form $\mathcal{E}_\varepsilon=\mathcal{E}_0+\varepsilon\mathcal{B}_1+O(\varepsilon^2)$ whose coefficients converge in L^∞. Testing the difference of its Euler equation and that for W against their zero trace difference gives an energy norm difference at most $C|\varepsilon|\|\nabla W\|_{L^2(y^a)}$. By the stationary Euler equation, replacing the minimizer by W in the energy changes it by $O(\varepsilon^2)$. Thus their first derivatives agree. All bilinear integrals are finite by the energy bound above.

Differentiate the change of variables in this transported energy. Since $\operatorname{div}X=\eta' r^{1-n}$ and the radial eigenvalue of DX is $\eta' r^{1-n}+(1-n)\eta r^{-n}$, radial integration gives

$$\frac{\delta\mathcal{E}}{|S^{n-1}|}=\int_0^\infty\left[-T_*(r)\eta'(r)+(n-1)\frac{A_*(r)}{r}\eta(r)\right]\mathrm{d}r.$$

Let η rise from zero to one across h, and then cut it off between L and $2L$. The first transition converges to $-T_*(h)$, because $h<1$ is in a pure phase. The outer contribution tends to zero: its absolute value is bounded by $CL^{-n}\int_L^{2L}(|T_*|+A_*)r^{n-1}\,\mathrm{d}r$, which tends to zero by the finite total extension energy. The remaining integral converges by positivity to the asserted radial expression. Its near-boundary integrability follows also from the energy bound since r is bounded above and below there.

To evaluate the same variation using the trace energy, put

$$H_\beta(B,e_1)=\mathrm{PV}\int_{\mathbb{R}^n}\frac{\mathbf{1}_{B^c}(t)-\mathbf{1}_B(t)}{|e_1-t|^{n+\beta}}\,\mathrm{d}t.$$

The first variation of Per_β at the ball is $|S^{n-1}|H_\beta(B,e_1)$, since $X\cdot\nu=1$ at its boundary. This formula may be obtained without multiplying a delta distribution by a singular trace. Replace the kernel by $(|x-t|^2+\delta^2)^{-(n+\beta)/2}$. The ordinary moving-domain formula for this smooth kernel gives the boundary integral of the difference of its integrals over B^c and B. In the pullback double integral the kernel derivative is bounded by a constant times the original cross kernel, since X is Lipschitz and bounded, so dominated convergence passes the derivative to the unregularized

energy. At the ball the regularized boundary difference converges to the principal value curvature: pair opposite rays from a boundary point; the signed near parts cancel and the remaining ray integrals are bounded by $C|\omega_1|^{-\beta}$, uniformly in the regularization parameter. This angular majorant is integrable because $\beta < 1$. Thus the radial stress equals $2C_{n,s}H_\beta(B, e_1)$.

For completeness, that ray pairing gives an elementary exact evaluation. Write $t = e_1 + \rho\omega$. Along the inward ray the interval inside B has length $2|\omega_1|$. Pair it with the opposite outward ray; the remaining contribution is $2\int_{2|\omega_1|}^\infty \rho^{-1-\beta}\,\mathrm{d}\rho$. Hence

$$H_\beta(B, e_1) = \frac{2^{-\beta}}{\beta}\int_{S^{n-1}} |\omega_1|^{-\beta}\,\mathrm{d}\omega = \frac{2^{1-\beta}\pi^{(n-1)/2}\Gamma((1-\beta)/2)}{\beta\Gamma((n-\beta)/2)}.$$

The sphere integral is the elementary beta integral. Substitute the constant $C_{n,s}$ and $|\Gamma(-s)| = \Gamma(1-s)/s$ to obtain the stated $\kappa_{n,s}$. This proves the normalization with no hidden step-function product. □

Lemma 7.5 (The sharp force for $0 < s < 1/2$)**.** *Under the assumptions of this section, for* $0 < s < 1/2$,

$$R(z)^{2s}\{(n-1)I_h(z) - T(h(z), z) + h'(z)P(h(z), z)\} \longrightarrow \kappa_{n,s}.$$

Proof. Use Lemma 7.1, Lemma 7.3 and Lemma 7.4. Put $R = R(z)$. Since $R' = o(1)$ and $R = o(z)$,

$$R(z + RT)/R \to 1$$

uniformly for bounded T. The whole-slice phase tails show that $U(RX, z + RT)$ tends almost everywhere to $2\mathbf{1}_{|X|<1} - 1$. Set $W_z(X, T, \eta) = V(RX, z + RT, R\eta)$. Differentiated Poisson kernels are integrable at every positive scaled extension height. Thus W_z and its first derivatives converge on compact positive-height sets to $W = P_\eta^{(n)} * (2\mathbf{1}_{B_1} - 1)$, independent of T. Writing $\rho = |X|$ and using the radial representative of W_z, the exact energy scalings are

$$R^\beta I_h(z) = d_s\int_0^\infty \eta^{1-\beta}\int_{1/2}^\infty \frac{(W_z)_\rho(\rho, 0, \eta)^2}{\rho}\,\mathrm{d}\rho\,\mathrm{d}\eta,$$

$$\begin{aligned} R^\beta T(h, z) = {} & \frac{d_s}{2}\int_0^\infty \eta^{1-\beta}((W_z)_\rho^2 - (W_z)_T^2 - (W_z)_\eta^2)(1/2, 0, \eta)\,\mathrm{d}\eta \\ & - R^\beta F(U(R/2, z)). \end{aligned}$$

We give tightness estimates before using this convergence. Radial variation and universal derivative bounds give

$$\int_0^\infty V_r^2\,\mathrm{d}r \le C\min(1, y^{-1}).$$

Since $r \ge R/2$ in I_h, for $0 < \varepsilon < 1$,

$$R^\beta\int_0^{\varepsilon R} y^{1-\beta}\int_{R/2}^\infty \frac{V_r^2}{r}\,\mathrm{d}r\,\mathrm{d}y \le CR^{\beta-1} + C\varepsilon^{1-\beta}.$$

The first term vanishes because $\beta < 1$. The radial symmetry estimate in Lemma 7.3 gives the upper-height tail

$$R^\beta\int_{MR}^\infty y^{1-\beta}\int_{R/2}^\infty \frac{V_r^2}{r}\,\mathrm{d}r\,\mathrm{d}y \le CM^{-\beta}.$$

On a fixed positive scaled height interval, the radial tail $r > KR$ is at most C/K after scaling, by $1/r \le 1/(KR)$, the variation bound, and $|V_r| \le C/y$. Compact convergence, followed by these three tightness bounds, proves

$$R^\beta I_h(z) \to \int_{1/2}^\infty A_*(r)/r\,\mathrm{d}r.$$

At $(R/2, z)$ a ball of radius γR, for fixed small $\gamma > 0$, stays a fixed fraction of R from the interface. Use the cutoff construction in the proof of Lemma 5.1 there, with $D = \gamma R/4$, to split $1 - U$ into near and remote parts. The near part has global C^2 norm $\delta = CR^{-\beta}$. Apply that lemma to this part with $K = \delta$ and an arbitrarily large ball, then let its radius tend to infinity.

Its complete slice energy is at most $C\delta^2$, negligible after multiplication by R^β. The remote part is bounded and vanishes in a ball of radius bR about the evaluation point, for a fixed $b > 0$. Denote its Poisson extension by V_{rem}. The remote-kernel estimates in the same slice proof, rescaled by R, give for $0 < y < bR/2$

$$|(V_{\mathrm{rem}})_y| \le Cy^{\beta-1}R^{-\beta}, \qquad |\nabla_{x,z}V_{\mathrm{rem}}| \le Cy^\beta R^{-1-\beta}.$$

For all $y > 0$, boundedness of the remote trace also gives $|\nabla V_{\mathrm{rem}}| \le C/y$. Consequently its complete scaled slice energy is $O(1)$, with the part below εR bounded by $C(\varepsilon^\beta + \varepsilon^{2+\beta})$ for $\varepsilon < b/2$, and the part above MR bounded by $CM^{-\beta}$. The mixed terms are bounded by the product of the square roots of the scaled near and remote energies. The first factor tends to zero and the second remains bounded, so these terms vanish by Cauchy–Schwarz. The potential contribution vanishes after scaling because $F(U(R/2, z)) = O(R^{-2\beta})$. These estimates and the compact convergence prove $R^\beta T(h, z) \to T_*(1/2)$. Finally $h' = o(1)$ and $P(h) = O(R^{-\beta})$. Combining the limits with Lemma 7.4 proves the assertion. □

7.4. **The critical mesoscopic energy.**

Lemma 7.6 (The logarithmic force at $s = 1/2$)**.** *Under the assumptions of this section, at* $s = 1/2$,

$$I_h(z) \sim \frac{2}{\pi}\frac{\log R(z)}{R(z)},$$

and therefore

$$(n-1)I_h - T(h, z) + h'P(h, z) \sim \frac{2(n-1)}{\pi}\frac{\log R}{R}.$$

Proof. Use Lemma 7.1 and Lemma 7.3. Here $a = 0$ and $d_{1/2} = 1$. We first prove the uniform mesoscopic assertion: for every sequence $z_j \to \infty$ and scales $y_j \to \infty$ with $y_j/R_j \to 0$, where $R_j = R(z_j)$,

$$R_j y_j \int_{R_j/2}^\infty \frac{V_r(r, z_j, y_j)^2}{r}\,\mathrm{d}r \longrightarrow \frac{2}{\pi}.$$

Rescale the trace around $(R_j e_1, z_j)$ by y_j. The angular curvature error divided by y_j is $O(y_j/R_j)$ on each compact set. Also $z_j - Cy_j \to \infty$ because $y_j = o(R_j) = o(z_j)$, and

$$\frac{|R(z_j + y_j T) - R_j|}{y_j} \le |T| \sup_{|t-z_j|\le Cy_j} R'(t) \to 0.$$

Together with the angular estimate this gives, uniformly on compact sets,

$$\frac{|R_j e_1 + y_j X| - R(z_j + y_j T)}{y_j} \longrightarrow X_1.$$

For $X_1 \neq 0$, the radial offset has the sign of X_1 and its absolute value tends to infinity. Thus the phase tails give convergence of the rescaled trace almost everywhere to $\mathrm{sgn}(-X_1)$. Normalized dimension reduction and dominated convergence of the differentiated Poisson kernels yield, uniformly on each fixed compact t interval,

$$y_j V_r(R_j + y_j t, z_j, y_j) \longrightarrow -\frac{2}{\pi(1+t^2)}.$$

The extension value at these points tends to $-2\arctan(t)/\pi$. The radial variation outside $|r - R_j| \le Ky_j$ is bounded as follows. The inner variation is $V(0, z_j, y_j) - V(R_j - Ky_j, z_j, y_j) \le 1 - V(R_j - Ky_j, z_j, y_j)$, and the outer variation is $V(R_j + Ky_j, z_j, y_j) + 1$. Thus the sum is bounded by

$$1 - V(R_j - Ky_j, z_j, y_j) + V(R_j + Ky_j, z_j, y_j) + 1.$$

Its limit is $2 - 4\arctan(K)/\pi$, which tends to zero as $K \to \infty$. Using $|V_r| \le C/y_j$ converts this variation into tightness of $y_j \int V_r^2\,\mathrm{d}r$. On the cut region $r \ge R_j/2$ one has $R_j/r \le 2$. Every fixed

window $|r - R_j| \le Ky_j$ lies in that region for large j, and $R_j/r \to 1$ uniformly on the window. Therefore the weighted energy has the same limit as

$$\int_{\mathbb{R}} \frac{4\,\mathrm{d}t}{\pi^2(1+t^2)^2} = \frac{2}{\pi}.$$

The sequential assertion implies the following uniform version. Given $\varepsilon > 0$, one can choose M large, δ small, and Z large so that for $z \ge Z$ and $M \le y \le \delta R(z)$,

$$\left| Ry \int_{R/2}^{\infty} V_r^2/r\,\mathrm{d}r - 2/\pi \right| \le \varepsilon.$$

Otherwise choose successively $M = j$, $\delta = 1/j$, and $Z = j$, to obtain a sequence contradicting the preceding limit. Empty scale intervals need no assertion. For sufficiently large z they are nonempty because $R \to \infty$. Integrate this uniform estimate in dy/y. The lower-height part $0 < y < M$ contributes $O_M(R^{-1})$ by derivative and variation bounds. The upper part $y > \delta R$ contributes $O(\delta^{-1}R^{-1})$ by $\int V_r^2/r \le C/y^2$. For fixed M, δ, both are negligible relative to $\log R/R$. The middle part is between $(2/\pi - \varepsilon)\log(\delta R/M)/R$ and $(2/\pi + \varepsilon)\log(\delta R/M)/R$. Let $z \to \infty$ and then $\varepsilon \downarrow 0$. This proves the equivalent for I_h. Finally $T(h) = O(R^{-1})$ and $h'P(h) = o(R^{-1})$ by Lemma 7.3, proving the force equivalent. □

7.5. **The height laws.** Lemma 4.4 gives $R_\theta/R \to 1$ and (4.10), with $H_\theta(r) \to \infty$. We use these facts for both height laws.

Proof of Theorem 1.4. Use Lemma 7.1, Lemma 7.2, Lemma 7.3, and Lemma 7.5. Set $S(z) = \int_Z^z R(t)^{-\beta}\,\mathrm{d}t$. Monotonicity gives $S \ge (z - Z)R(z)^{-\beta}$, so $R = o(z)$ implies $z^{1-\beta} = o(S)$ and $S \to \infty$. Thus the renormalized flux is $o(S)$. Integrating the force equivalent by the fixed-height error split used in the proof of Theorem 1.5 gives

$$\int_Z^z \{(n-1)I_h - T(h,t) + h'P(h,t)\}\,\mathrm{d}t = \kappa_{n,s}S(z) + o(S(z)).$$

Stress and drift therefore give

$$(cm_s + o(1))R = (\kappa_{n,s} + o(1))S, \qquad \frac{R}{S} \longrightarrow A = \frac{\kappa_{n,s}}{cm_s} > 0.$$

Lemma 5.4 with $\gamma = \beta$ gives

$$\frac{R(z)^{1+\beta}}{z} \longrightarrow (1+\beta)A.$$

The fixed-level ratio and inversion above yield

$$\frac{H_\theta(r)}{r^{1+2s}} \longrightarrow \frac{1}{(1+2s)A} = \frac{cm_s}{(1+2s)\kappa_{n,s}}.$$ □

Proof of Theorem 1.3. Use Lemma 7.1, Lemma 7.2, Lemma 7.3 and Lemma 7.6. Put $a_0 = 2(n-1)/\pi$ and $S(z) = \int_Z^z \log R(t)/R(t)\,\mathrm{d}t$, taking Z so that $R > e$. Since $R = o(z)$, $\int_Z^z R^{-1}\,\mathrm{d}t/\log z \to \infty$. As $\log R \ge 1$, the same holds for S, and the logarithmic flux bound is $o(S)$. The same fixed-height error split integrates the force equivalent with weight $\log R/R$. Stress and drift give

$$(cm_{1/2} + o(1))R = (a_0 + o(1))S, \qquad R/S \to A = a_0/(cm_{1/2}) > 0.$$

In particular $\log R/\log S \to 1$. Let $\Phi(v) = \int_{v_0}^v t/\log t\,\mathrm{d}t$ for $v \ge v_0 > 1$. Then $\Phi(v) \sim v^2/(2\log v)$ by l'Hôpital's rule. At all large z, $S(z) > v_0$, and

$$\frac{d}{dz}\Phi(S(z)) = \frac{S}{\log S}\frac{\log R}{R} \longrightarrow A^{-1}.$$

Integration gives $\Phi(S(z))/z \to A^{-1}$, hence $S^2/(2z\log S) \to A^{-1}$. Combining this with $R/S \to A$ and $\log R/\log S \to 1$ yields $R^2/(2z\log R) \to A$. Every fixed level has $R_\theta/R \to 1$ and consequently $\log R_\theta/\log R \to 1$. The common inversion above now gives

$$\frac{H_\theta(r)\log r}{r^2} \longrightarrow \frac{1}{2A} = \frac{\pi cm_{1/2}}{4(n-1)}.$$ □

Appendix A. Interior and directional boundary estimates

A.1. **Interior compactness with constant drift.**

Lemma A.1 (Interior estimates)**.** *Let $d \geq 1$, $0 < s < 1$, $b \in \mathbb{R}^d$, and let D be open. A bounded distributional solution w of*

$$L_{d,s}w - b \cdot \nabla w + f(w) = 0 \quad \text{in } D$$

with $f \in C^2$ on a neighborhood of its range belongs locally to $C^{2,\eta}(D)$ for every $0 < \eta < \min\{2s, 1\}$. On $K \Subset D$, the estimate depends only on $d, s, \eta, |b|$, the distance to D^c, the global bound for w, and the C^2 norm of f on its range. These estimates are global when $D = \mathbb{R}^d$. For entire solutions with $s > 1/2$, one also has a global $C^{3,\gamma}$ estimate for every $0 < \gamma < 2s - 1$. Consequently, uniformly bounded solutions with fixed equation data on exhausting domains are locally precompact in C^2.

For $s > 1/2$, the following linear estimates hold on fixed nested balls. Write

$$T(w; X_0) = \int_{\mathbb{R}^d} \frac{|w(Y)|}{(1 + |Y - X_0|)^{d+2s}} \, \mathrm{d}Y.$$

If $w \in L^\infty(B_2(X_0))$, $T(w; X_0) < \infty$, and $(L_{d,s} - c\partial_z)w = h$ distributionally on $B_2(X_0)$ with $h \in L^\infty(B_2(X_0))$, then

$$\|w\|_{C^{1,\gamma}(B_1(X_0))} \leq C\big(\|w\|_{L^\infty(B_2(X_0))} + T(w; X_0) + \|h\|_{L^\infty(B_2(X_0))}\big) \qquad (0 < \gamma < 2s - 1). \tag{A.1}$$

If $h = -a(X)w$ and a has a bounded C^1 norm, then

$$\|w\|_{C^2(B_1(X_0))} \leq C\big(\|w\|_{L^\infty(B_2(X_0))} + T(w; X_0)\big). \tag{A.2}$$

The constants are uniform for bounded c; in (A.2) they also depend on the $C^1(B_2(X_0))$ bound for a. Local C^2 forcing may be added to the semilinear equation, with its local norms included in the estimate.

Proof. The spatial estimates in [6, Theorem 1.3 and Corollary 3.8] give C^θ regularity for noninteger $\theta < 2s$ from bounded forcing, and for $\theta < \beta + 2s$ from spatially C^β forcing. The fractional Laplacian has the required smooth angular kernel. We use only spatial estimates with source norms uniform in time: the positive loss in Corollary 3.8 is chosen below the strict exponent gap, and each source exponent avoids the finitely many values for which $\beta + 2s$ is an integer. No temporal exterior regularity is needed.

The distributional change of variables $u(t, X) = w(X - bt)$ gives

$$\partial_t u + L_{d,s}u = -f(u) \quad \text{where } X - bt \in D.$$

For $K \Subset D$, choose $\rho > 0$ with $2\rho + 2|b|\rho^{2s} < \operatorname{dist}(K, D^c)/2$. Thus cylinders of radius 2ρ and times $(-2\rho^{2s}, 2\rho^{2s})$, centered at points of K, stay in this moving domain. This choice works at every $s > 0$. To start from distributions, spatially mollify w on a smaller set. The transformed smooth functions have uniformly bounded source $-\varphi_\delta * f(w)$. Testing with compact cutoffs $\zeta^2(t)\xi^2(X)u_\delta$ and using

$$(a - b)(\lambda^2 a - \mu^2 b) = (\lambda a - \mu b)^2 - ab(\lambda - \mu)^2$$

gives a uniform local space–time H^s bound: the cutoff error is integrable because $s < 1$. More precisely, for fixed compact cutoffs ζ, ξ supported in the chosen cylinder,

$$\int \zeta(t)^2 \|\xi u_\delta(t)\|^2_{H^s(\mathbb{R}^d)} \, \mathrm{d}t \leq C(\|w\|^2_\infty + \|w\|_\infty \|f(w)\|_\infty),$$

with C independent of δ. Local strong L^2 convergence of u_δ and its source, together with weak compactness in this energy space, passes the localized weak formulation; boundedness controls its nonlocal tail. Thus the transformed limit is a bounded weak solution to which the cited spatial estimates apply, yielding a continuous representative.

Bootstrap on nested cylinders. Composition with $f \in C^2$ preserves every noninteger C^β norm with $\beta < 2$: use Lipschitz continuity for $\beta < 1$, and $\nabla(f(w)) = f'(w)\nabla w$ with bounded f'' for $1 < \beta < 2$. Given $\eta < \min\{2s, 1\}$, choose a noninteger $\beta_* < 2$ with $2 + \eta < \beta_* + 2s$. Starting below $\min\{2s, 1\}$, finitely many gains smaller than s reach β_*; the final estimate gives $C^{2,\eta}$. All

exponent choices lie in open intervals, so the nonresonance conditions above cause no obstruction. For $s > 1/2$, bounded second derivatives imply

$$\partial_{ij}(f(w)) = f'(w)\partial_{ij}w + f''(w)\partial_i w\partial_j w,$$

so $f(w) \in C^{1,1}$ locally. Choosing $\beta < 2$ with $3 + \gamma < \beta + 2s$ gives $C^{3,\gamma}$ for every $\gamma < 2s - 1$, without a Hölder assumption on f''.

Additional $g \in C^2_{\mathrm{loc}}(D)$ contributes $g(X - bt)$ to the transformed source. If $K' \Subset D$ contains the spatial points swept out by these cylinders, then, for $0 < \beta < 2$ noninteger,

$$\sup_t \|g(\cdot - bt)\|_{C^\beta(B_{2\rho})} \le C_{\rho,\beta}\|g\|_{C^2(K')}.$$

The same bounds hold after mollification. They suffice at every bootstrap step; once $w \in C^2$, $g - f(w) \in C^{1,1}$ gives the high-order improvement as well. Global estimates require these local forcing norms uniformly under translation, for example a global C^2 bound for g.

For the tail-sensitive bounds translate $X_0 = 0$, choose $\chi = 1$ on $B_{3/2}$, supported in $B_{7/4}$, and set $v = \chi w$. On $B_{4/3}$,

$$(L_{d,s} - c\partial_z)v = h + E_\chi, \qquad E_\chi(X) = C_{d,s}\int_{\mathbb{R}^d} \frac{(1 - \chi(Y))w(Y)}{|X - Y|^{d+2s}}\,\mathrm{d}Y.$$

The fixed separation from the discarded data gives $\|D^j E_\chi\|_\infty \le C_j T(w; 0)$ for every j. The bounded-source estimate with $1 + \gamma < 2s$ proves (A.1). If $h = -aw$ and a is bounded in C^1, that estimate on an intermediate ball bounds aw in the Lipschitz norm by $C(\|w\|_{L^\infty(B_2)} + T(w; 0))$. A source exponent in $(2 - 2s, 1)$ gives (A.2).

For entire solutions translation gives uniform derivative bounds; local Hölder bounds control pairs at distance at most one and derivative bounds control the rest. Arzelà–Ascoli gives local C^2 compactness. The operator passes to the limit by second differences near zero, local convergence on annuli, and the uniform tail bound $C\|w\|_\infty R^{-2s}$ beyond radius R. □

A.2. **Shifted powers at the top and side.**

Lemma A.2 (Boundary powers for the ordered approximation)**.** *Let $0 < s < 1$, $c \ge 0$, $M > 0$, and $\Omega_R = B^n_R \times (-\infty, 0)$. Suppose that u is an L^1_{loc} limit of bounded capped-kernel Dirichlet approximations, with uniformly bounded forcing, for*

$$(L_{n+1,s} - c\partial_z + M)u = h \quad \text{in } \Omega_R.$$

The approximations use fixed bounded full exterior data g_{ext} and, when $c > 0$, the assigned top values $g_{\mathrm{ext}}(x, 0)$ in the transport formula. If $g_{\mathrm{ext}} = \ell$ near a smooth top or side point away from the junction, then the continuous interior representative of u satisfies, for each $0 < p < s$,

$$|u(X) - \ell| \le C\,\mathrm{dist}(X, \partial\Omega_R)^p \tag{A.3}$$

in a smaller neighborhood. The constants are uniform under translations preserving the local geometry, patch sizes and data bounds. The assertion also holds on the half-line. If $s > 1/2$, a bounded classical half-line solution $v = 0$ on $[0, \infty)$ of $L_{1,s}v - cv' = h$ with bounded h satisfies

$$|v(-r)| \le Cr^p, \qquad |v'(-r)| \le Cr^{p-1} \quad (0 < r < 1/2). \tag{A.4}$$

Proof. For $0 < p < s$, the flat power has a positive fractional Laplacian:

$$L_{1,s}(t^p_+) = k_{s,p}t^{p-2s} \quad (t > 0), \qquad \frac{k_{s,p}}{C_{1,s}} = \frac{1}{2s} + \mathrm{PV}\int_0^\infty \frac{1 - t^p}{|1 - t|^{1+2s}}\,\mathrm{d}t.$$

Indeed, this coefficient is strictly concave in $p \in (0, 2s)$, as is seen by differentiating twice under the integral. Its values at $p = 0$ and $p = s$ are $C_{1,s}/(2s)$ and 0, respectively; the latter is the half-space harmonic distance power [9, Proposition 3.1]. Hence $k_{s,p} > 0$. Cutting the power off away from its zero changes its operator by a bounded term.

For the interior distance ρ near a C^2 boundary, extend ρ^p_+ by zero outside the domain and truncate it boundedly away from the patch. The corresponding local calculation gives

$$L_{d,s}(\rho^p_+) = k_{s,p}\rho^{p-2s} + o(\rho^{p-2s}). \tag{A.5}$$

To verify the error, scale displacements by $\rho(X)$. The rescaled boundary tends to its tangent hyperplane; dimension reduction (2.2) identifies its coefficient with $k_{s,p}$. Around the evaluation point symmetric second differences are bounded by $C|Z|^{2-d-2s}$; away from that point the bound is $C(1+|Z|^p)|Z|^{-d-2s}$. Both are integrable, since $s<1$ and $p<2s$. Dominated convergence is uniform on a smaller chart of bounded C^2 geometry. This is the distance power calculation underlying [9].

At the top take a nonnegative bounded truncation of $(-z)_+^p$. With $\rho=-z$, the operator including drift and mass is

$$k_{s,p}\rho^{p-2s}+cp\rho^{p-1}+M\rho^p+O(1).$$

The drift term is favorable. At the side use $(R-|x|)_+^p$, independent of z; its drift is zero, and (A.5) applies after dimension reduction. Thus both patches admit a barrier ψ with a positive lower bound $a_0\rho^{p-2s}$ in a sufficiently thin strip, at every $s\in(0,1)$.

The capped approximations are uniformly bounded. Put $G=\|g_{\mathrm{ext}}\|_\infty$, including the assigned top values, $H=\sup_\varepsilon\|h_\varepsilon\|_\infty$, and $B_0=\max\{G,H/M\}$. For $c>0$, the proper comparison proved from (3.3) in Lemma 3.2, applied to the constant super- and subsolutions B_0 and $-B_0$, gives $\|u_\varepsilon\|_\infty\le B_0$. For $c=0$, the map

$$\mathcal{T}_\varepsilon v=\frac{h_\varepsilon+k_\varepsilon*\operatorname{Ext} v}{M+\lambda_\varepsilon}$$

preserves the ball $\|v\|_\infty\le B_0$, since $H+\lambda_\varepsilon B_0\le(M+\lambda_\varepsilon)B_0$, and has contraction factor $\lambda_\varepsilon/(M+\lambda_\varepsilon)<1$. Its bounded fixed point has the same bound. These arguments apply on the cylinder and the half-line.

We establish the estimate before passing to the singular kernel. Fix a patch ball $B_{2r_0}(X_0)$ meeting only the chosen boundary component, with exterior data ℓ throughout the patch. Let $\chi\ge0$ be smooth, vanish on $B_{r_0/2}(X_0)$ and equal one outside $B_{3r_0/4}(X_0)$. Let

$$S=\Omega_R\cap B_{r_0}(X_0)\cap\{0<\rho<\delta\}.$$

The cutoff has uniformly bounded operator. Its coefficient will control data away from the patch, whereas the coefficient of the power will control the forcing and the artificial boundary $\rho=\delta$.

Write $L_\varepsilon=\lambda_\varepsilon I-k_\varepsilon*$ for the capped operator in Lemma 3.2, and take $\eta=\sqrt{\varepsilon}$, with ε small enough that $2\varepsilon<\eta<\delta<r_0$. At the top replace the power by $\psi_\eta=\psi_0(-z+\eta)$, where ψ_0 is a fixed nonnegative bounded truncation agreeing with t_+^p for $0\le t\le 2r_0$. At the side take $\psi_\eta=(R+\eta-|x|)_+^p$. Near the patch, set $t=\rho(X)+\eta$. The shifted surfaces have uniformly bounded geometry. Decreasing δ if necessary, their full-operator lower bound is a_0t^{p-2s} for $0<t<2\delta$. Since $t\ge\eta>2\varepsilon$, the Hessian bound and the consistency estimate (3.4) give

$$\sup_{B_\varepsilon(X)}|D^2\psi_\eta|\le Ct^{p-2},\qquad |(L_{d,s}-L_\varepsilon)\psi_\eta(X)|\le C\varepsilon^{2-2s}t^{p-2}\le C(\varepsilon/\eta)^{2-2s}t^{p-2s}.$$

The last factor tends to zero. The same bound for a fixed smooth cutoff is uniform in ε. Choose first B to control the bounded data off the patch and then A large enough to control the forcing, the uniformly bounded negative part of $B(L_\varepsilon-c\partial_z+M)\chi$, and the inner artificial boundary $\rho=\delta$. These choices, independent of ε, give

$$\begin{gathered}\Phi_\eta=A\psi_\eta+B\chi,\\ (L_\varepsilon-c\partial_z+M)\Phi_\eta\ge H+M|\ell|\ge|h_\varepsilon-M\ell|\quad\text{in } S,\\ \Phi_\eta\ge|u_\varepsilon-\ell|\quad\text{on } S^c.\end{gathered}$$

Here $\psi_\eta\ge\rho^p$ on the inner patch and is nonnegative on the exterior. Its nonzero value at the original boundary causes no problem.

For either sign let $w=\pm(u_\varepsilon-\ell)-\Phi_\eta$. Then, with $\kappa_\varepsilon=\lambda_\varepsilon+M$,

$$-c\partial_z w+\kappa_\varepsilon w\le k_\varepsilon*w\text{ in }S,\qquad w\le0\text{ on }S^c.$$

When $c>0$, take the vertical component (a_x,b_x) of S containing (x,z). Its upper endpoint is finite. The transport formula for the capped solution gives an absolutely continuous representative on this interval. At an artificial upper endpoint its interior limit agrees with the adjacent

representative; at the top it has the assigned endpoint value. Thus $w(x,b_x^-)\le 0$. Integration of the scalar inequality gives

$$\begin{aligned} w(x,z) &\le e^{-\kappa_\varepsilon(b_x-z)/c}w(x,b_x^-)+\frac{1}{c}\int_z^{b_x} e^{-\kappa_\varepsilon(t-z)/c}(k_\varepsilon * w)(x,t)\,\mathrm{d}t \\ &\le \frac{\lambda_\varepsilon}{\lambda_\varepsilon+M}\|w_+\|_{L^\infty(S)}. \end{aligned}$$

Taking positive suprema proves $w\le 0$. For $c=0$ the pointwise algebraic inequality gives the same contraction. The argument also applies to essential suprema.

Both signs therefore yield $|u_\varepsilon-\ell|\le\Phi_\eta$. The powers converge uniformly as $\eta\downarrow 0$. Pass along an almost-everywhere convergent subsequence of the L^1_{loc} Dirichlet limit. The bounded-source step in the proof of Lemma A.1, applied to $L_{n+1,s}u-c\partial_z u=h-Mu$, gives an interior continuous representative, so the estimate holds everywhere inside. On the smaller ball $\chi=0$, which proves the trace as well. The construction is uniform along translated side patches at heights $z\le -1$ for fixed R. A half-line uses the same top argument.

It remains to prove the last assertion for an arbitrary bounded classical half-line solution; no capped-approximation or boundary-continuity hypothesis is imposed on this solution. Put $\beta=2s>1$ and fix $0<p<s$. The flat-power formula above extends to every exponent $\gamma\in(-1,\beta)$, with the negative power defined to be zero on $[0,\infty)$ after reflection $t=-z$. The coefficient $k_{s,\gamma}$ is continuous at $\gamma=0$. Indeed, choose $q_0\in(0,\min\{1,\beta\})$ and restrict $|\gamma|\le q_0$. Near $t=1$, subtract the odd linear term $-\gamma(t-1)$ using an even cutoff about 1; its principal-value integral vanishes, and the remaining numerator is bounded by $C|t-1|^2$, uniformly in γ. Near zero and infinity the integrable majorants are respectively $C(1+t^{-q_0})$ and $C(1+t^{q_0})t^{-1-\beta}$. Dominated convergence therefore gives continuity. Since $k_{s,0}=C_{1,s}/\beta>0$, choose $q\in(0,1)$ small enough that $k_{s,-q}>0$.

Let $\psi_1(z)=(-z)^{-q}$ for $z<0$ and $\psi_1=0$ for $z\ge 0$. For $r=-z>0$ and $A_M=L_{1,s}-c\partial_z+M$,

$$A_M\psi_1(-r)=k_{s,-q}r^{-q-\beta}-cqr^{-q-1}+Mr^{-q}.$$

Because $\beta>1$, this is nonnegative for $0<r<\delta$ once $\delta>0$ is sufficiently small. Let ψ_0 be a nonnegative bounded truncation of $(-z)_+^p$, equal to $(-z)^p$ for $0<-z<1/2$, positive on the negative half-line, and a positive constant for $-z\ge 1$. Choose its transition smooth away from zero. Its difference from the untruncated power changes the fractional Laplacian by a bounded amount for $0<r<1/4$, so

$$A_M\psi_0(-r)=k_{s,p}r^{p-\beta}+cpr^{p-1}+Mr^p+O(1).$$

After decreasing $\delta<1/4$, we have $A_M\psi_0\ge a_0r^{p-\beta}$ on $(-\delta,0)$ for some $a_0>0$, and $\inf_{z\le-\delta}\psi_0(z)>0$.

Since $A_Mv=h+Mv$, choose $A>0$, independently of $\varepsilon>0$, such that

$$Aa_0\delta^{p-\beta}\ge\|h\|_\infty+M\|v\|_\infty,\qquad A\inf_{z\le-\delta}\psi_0(z)\ge\|v\|_\infty.$$

For either sign set $w=\pm v-A\psi_0-\varepsilon\psi_1$. Then $A_Mw\le 0$ on $(-\delta,0)$, $w\le 0$ outside that interval, and $w(z)\to-\infty$ as $z\uparrow 0$. If w were positive somewhere, its positive global maximum would therefore be attained at an interior point $z_*\in(-\delta,0)$. At that point $w'(z_*)=0$ and $L_{1,s}w(z_*)\ge 0$, whereas $Mw(z_*)>0$, contradicting $A_Mw(z_*)\le 0$. The singular integral used here is well defined: w is C^2 near z_*, the only added boundary singularity is integrable because $q<1$, and its tail is bounded. Letting $\varepsilon\downarrow 0$ gives $|v(-r)|\le Ar^p$ for $0<r<\delta$. Enlarging the constant using boundedness proves this estimate for $0<r<1/2$.

For the derivative estimate, take $0<r<\delta/4$ and set $v_r(t)=r^{-p}v(-r+rt)$. On $t<1$ its equation is

$$L_{1,s}v_r-cr^{\beta-1}v_r'=r^{\beta-p}h(-r+rt).$$

The power bound and boundedness imply $|v_r(t)|\le C(1+|t|)^p$ on the full real line, with C independent of r: use the power estimate when $r(1-t)<\delta$ and boundedness when $r(1-t)\ge\delta$; for $t\ge 1$ the function is zero. Its weighted tail is uniformly finite because $p<\beta$, and the drift and

source bounds are uniform because $\beta > 1$ and $\beta > p$. Apply (A.1) on fixed nested intervals about $t = 0$ contained in $(-1/2, 1/2)$. It gives $|v_r'(0)| \le C$, hence $|v'(-r)| \le Cr^{p-1}$. For $\delta/4 \le r < 1/2$, the same interior estimate with fixed positive distance from zero, the global bound for v, and the bound for h gives a uniform derivative bound. This proves (A.4) for the full stated range. □

A.3. **The critical half-line rescaling.**

Lemma A.3 (Critical half-line modulus)**.** *Let $c \ge 0$, and suppose that $v \ge 0$ is bounded, $v = 0$ on $[0, \infty)$, and solves in distributions*

$$L_{1,1/2} v - cv' = f(1 - v) \quad \text{on } (-\infty, 0).$$

If $f \in C^2$, $f(1) = 0$, and $v(-r) \le C_0 r^p$ for $0 < r \le 1$ and some $0 < p < 1$, then v has a representative in $C^{0,p}(\mathbb{R})$ satisfying $|v'(-r)| \le Cr^{p-1}$ near zero. The constants depend only on the displayed data and the global bound for v.

Proof. For $0 < r \le 1/2$, put $V_r(t) = r^{-p} v(rt)$. The power estimate and boundedness give

$$|V_r(t)| \le C(-t)_+^p, \qquad \int_{\mathbb{R}} \frac{|V_r(t)|}{1 + t^2} \, \mathrm{d}t \le C, \qquad L_{1,1/2} V_r - cV_r' = G_r(V_r),$$

where $G_r(a) = r^{1-p} f(1 - r^p a)$ has uniformly bounded C^2 norms on bounded ranges: $G_r(0) = 0$, $G_r' = -rf'(1 - r^p a)$, and $G_r'' = r^{1+p} f''(1 - r^p a)$. Choose a smooth cutoff $0 \le \chi \le 1$ supported in $(-2, -1/4)$ and equal to one near $[-5/4, -3/4]$. Then $Z_r = \chi V_r$ is globally bounded and near -1 satisfies

$$L_{1,1/2} Z_r - cZ_r' = G_r(Z_r) + E_r, \qquad E_r(t) = C_{1,1/2} \int_{\mathbb{R}} \frac{(1 - \chi(\tau)) V_r(\tau)}{|t - \tau|^2} \, \mathrm{d}\tau.$$

The separation from discarded data and the weighted tail bound control every local derivative of E_r uniformly. Apply Lemma A.1 directly to Z_r, with reaction $-G_r$ and local forcing E_r. Its global bound and the local C^2 source norms are independent of r, so $|V_r'(-1)| \le C$. Hence the interior representative satisfies $|v'(-r)| \le Cr^{p-1}$ for $0 < r \le 1/2$. Translated interior estimates for the original equation give a uniform derivative bound on $(-\infty, -1/2]$. They also extend the power derivative bound to $0 < r < 1$. The assumed value bound makes this representative continuous at zero, where its exterior value is zero. Lemma 2.3 now gives $v \in C^{0,p}(\mathbb{R})$. □

Appendix B. The positive massive resolvent

Lemma B.1 (Positive massive resolvent)**.** *For $d \ge 1$, $0 < s < 1$, $c \ge 0$, and $\lambda > 0$, set $A_\lambda = L_{d,s} - c\partial_z + \lambda$. There is a smooth, strictly positive kernel p_t, $t > 0$, with $\widehat{p}_t(\xi) = e^{-t|\xi|^{2s}}$ and mass one. For bounded h, the unique bounded distributional solution of $A_\lambda w = h$ is*

$$w(X) = \int_0^\infty e^{-\lambda t} (p_t * h)(X + cte_z) \, \mathrm{d}t = G_\lambda * h(X), \qquad G_\lambda(Y) = \int_0^\infty e^{-\lambda t} p_t(Y + cte_z) \, \mathrm{d}t. \tag{B.1}$$

The time integral defining G_λ is finite away from the origin and determines an L^1 kernel; its value at the origin may be infinite and does not affect convolution. Moreover, $G_\lambda > 0$ almost everywhere, $\int G_\lambda = 1/\lambda$, and

$$G_\lambda(Y) \le C|Y|^{-d-2s} \qquad (|Y| \ge 1). \tag{B.2}$$

Consequently,

$$\int_{|Y| \ge D} G_\lambda(Y) \, \mathrm{d}Y \le CD^{-2s} \qquad (D \ge 1). \tag{B.3}$$

Here C depends only on d, s, c, λ. For bounded w, h, the distributional inequality $A_\lambda w \le h$ implies $w \le G_\lambda * h$ almost everywhere, and everywhere if w is continuous.

Proof. Let $\nu = 2s \in (0, 2)$ and use $\widehat{v}(\xi) = \int e^{-iX \cdot \xi} v(X) \, \mathrm{d}X$. Define

$$p_t(X) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{iX \cdot \xi - t|\xi|^\nu} \, \mathrm{d}\xi.$$

This integral and all differentiated integrals converge for $t > 0$, and

$$p_t(X) = t^{-d/\nu} p_1(t^{-1/\nu} X). \tag{B.4}$$

We first obtain integrable estimates directly from the multiplier, so that no sign property is being used in the construction.

Choose a radial smooth cutoff χ equal to one on B_1 and zero outside B_2, and put $\beta = \chi - \chi(2\,\cdot)$. For $m(\xi) = e^{-|\xi|^\nu}$,

$$m = \chi + (1-\chi)m + \sum_{j\ge 0} \beta(2^j \xi)(m-1).$$

The first two terms are Schwartz functions. On the fixed annulus supporting β, every prescribed finite collection of derivatives of $\beta(\eta)(e^{-2^{-j\nu}|\eta|^\nu} - 1)$ is bounded by $C_N 2^{-j\nu}$. Rescaling and integration by parts yield

$$\left|\mathcal{F}^{-1}[\beta(2^j\,\cdot)(m-1)](X)\right| \le C_N 2^{-j(d+\nu)} (1 + 2^{-j}|X|)^{-N}.$$

Take $N > d + \nu$ and split at $2^j = |X|$. For $2^j \le |X|$ the sum is bounded by

$$C_N |X|^{-N} \sum 2^{j(N-d-\nu)} \le C|X|^{-d-\nu};$$

the remaining sum has the same bound. For bounded X the series converges uniformly. Multiplying each symbol by $(i\xi)^\gamma$ adds the factor $2^{-j|\gamma|}$. Repeating the summation with $N > d + \nu + |\gamma|$ gives

$$|D^\gamma p_1(X)| \le C_\gamma (1 + |X|)^{-d-\nu-|\gamma|}. \tag{B.5}$$

The decomposition is valid first in tempered distributions; the uniform convergence just established identifies its inverse with the continuous integral above. Thus p_t and its spatial derivatives belong to L^1, Fourier inversion gives $\int p_t = 1$, and scaling gives $\|D^\gamma p_t\|_1 \le C_\gamma t^{-|\gamma|/\nu}$. Differentiating (B.4), with $Z = t^{-1/\nu} X$, gives

$$\partial_t p_t(X) = -\frac{t^{-1-d/\nu}}{\nu} \left[d\, p_1(Z) + Z \cdot \nabla p_1(Z) \right].$$

Both terms in brackets are integrable by (B.5), so $\|\partial_t p_t\|_1 \le C t^{-1}$.

To justify the heat equation, write $\delta_H^2 p_t(X) = p_t(X+H) + p_t(X-H) - 2p_t(X)$. The preceding bounds give

$$\|\delta_H^2 p_t\|_1 \le \min\{|H|^2 \|D^2 p_t\|_1, 4\|p_t\|_1\}.$$

This bound is integrable against $|H|^{-d-\nu}\,\mathrm{d}H$, since $0 < \nu < 2$. Thus $L_{d,s} p_t \in L^1$, and Fubini in the symmetric singular integral identifies its Fourier transform with $|\xi|^\nu e^{-t|\xi|^\nu}$. Fourier inversion yields

$$\partial_t p_t + L_{d,s} p_t = 0. \tag{B.6}$$

The identity holds classically: bounded second derivatives control small displacements and boundedness controls the tail, making the singular integral continuous. Its continuous representative agrees with the Fourier inverse above.

Let $\varphi \ge 0$ be smooth and compactly supported. The function $u(t) = p_t * \varphi$ lies in $C([0,T]; C_0(\mathbb{R}^d))$, with $u(0) = \varphi$. At zero this follows from (B.4), the integrable bound for $|p_1|$ and mass one; at positive times it follows from continuity of dilations in L^1. It solves the heat equation for $t > 0$. If it takes a negative value, then $u + \varepsilon t$ does so for sufficiently small ε. Continuity into C_0 on a compact time interval makes its negative minimum occur at finite X and positive time. There the time derivative is nonpositive, including the left derivative at the final time, and the fractional Laplacian is nonpositive. This contradicts the equation with right-hand side $\varepsilon > 0$. Hence $p_t * \varphi \ge 0$; approximation of a point mass proves $p_t \ge 0$.

If $p_{t_0}(X_0) = 0$, nonnegativity in a time neighborhood gives $\partial_t p_{t_0}(X_0) = 0$. Smoothness and nonnegativity give quadratic vanishing at X_0, while the unit mass gives

$$L_{d,s} p_{t_0}(X_0) = -C_{d,s} \int_{\mathbb{R}^d} \frac{p_{t_0}(Y)}{|X_0 - Y|^{d+2s}}\,\mathrm{d}Y < 0.$$

This contradicts (B.6), so positivity is strict.

Scaling and (B.5) give

$$p_t(X) \le \frac{Ct}{(t^{1/\nu} + |X|)^{d+\nu}}.$$

Integration, together with mass one, gives for $D, t > 0$

$$\int_{|X|\ge D} p_t(X)\,\mathrm{d}X \le C \min\{1, tD^{-2s}\}. \tag{B.7}$$

When $c = 0$, integrate $p_t(Y) \le Ct|Y|^{-d-\nu}$ against $e^{-\lambda t}$. When $c > 0$ and $|Y| \ge 1$, split at $T = |Y|/(2c)$. For $t \le T$, $|Y + cte_z| \ge |Y|/2$; for $t > T$, use $p_t \le Ct^{-d/\nu}$. Hence

$$G_\lambda(Y) \le C|Y|^{-d-\nu} \int_0^\infty te^{-\lambda t}\,\mathrm{d}t + C\int_T^\infty e^{-\lambda t}t^{-d/\nu}\,\mathrm{d}t \le C|Y|^{-d-\nu} + Ce^{-\lambda T/2} \le C|Y|^{-d-\nu}.$$

Here $T \ge 1/(2c)$, so the constants depend only on the stated data. This proves (B.2); integration in polar coordinates gives (B.3). The same time split proves finiteness at every $Y \ne 0$, with a constant allowed to depend on Y: near $t = 0$ the integrand is at most $C_Y t$, and the remaining time interval stays away from zero. Positivity and the mass of G_λ follow from strict positivity of p_t and Tonelli's theorem, so $G_\lambda \in L^1$. At $Y = 0$ the time integral can diverge: when $c = 0$ and $d \ge \nu$, scaling gives the nonintegrable factor $t^{-d/\nu}p_1(0)$ near zero. Its value there is immaterial to the L^1 kernel.

Set $T_t h(X) = (p_t * h)(X + cte_z)$. The heat equation implies $\partial_t T_t h = -(L_{d,s} - c\partial_z)T_t h$ in distributions. The scaling, integrability and mass of p_t make it an approximate identity, so $T_t h \to h$ as $t \downarrow 0$ in distributions. Also $\|T_t h\|_\infty \le \|h\|_\infty$. For $\varphi \in C_c^\infty(\mathbb{R}^d)$, set $A_\lambda^*\varphi = L_{d,s}\varphi + c\partial_z\varphi + \lambda\varphi$. It belongs to L^1 by the same second-difference bound used above. For $0 < a < b < \infty$, integration in time gives

$$\int_a^b e^{-\lambda t}\langle T_t h, A_\lambda^*\varphi\rangle\,\mathrm{d}t = e^{-\lambda a}\langle T_a h, \varphi\rangle - e^{-\lambda b}\langle T_b h, \varphi\rangle.$$

The uniform bound for $T_t h$ permits $a \downarrow 0$, $b \uparrow \infty$, and proves $A_\lambda w = h$ for the bounded function in (B.1). The equality $w = G_\lambda * h$ follows by Fubini, since the absolute integral is at most $\|h\|_\infty/\lambda$. Independently, Fubini for $G_\lambda \in L^1$ gives

$$\widehat{G}_\lambda(\xi) = \int_0^\infty e^{-t(\lambda + |\xi|^{2s} - ic\xi_z)}\,\mathrm{d}t = (|\xi|^{2s} - ic\xi_z + \lambda)^{-1}.$$

It remains to prove comparison and uniqueness in the stated bounded distributional class. Mollify $w - G_\lambda * h$ with a nonnegative smooth compactly supported kernel. The resulting smooth bounded function v has bounded derivatives and satisfies $A_\lambda v \le 0$ pointwise. Indeed, mollification commutes with A_λ, preserves the distributional inequality, and makes $A_\lambda v$ continuous. The same symmetric second-difference bound, now in the supremum norm, permits Fubini and gives $A_\lambda T_t v = T_t A_\lambda v$. Hence the heat equation and positivity give

$$\frac{\mathrm{d}}{\mathrm{d}t}(e^{-\lambda t}T_t v) = -e^{-\lambda t}T_t(A_\lambda v) \ge 0.$$

As $t \downarrow 0$, the left-hand function converges to v uniformly, since v is uniformly continuous. As $t \to \infty$ it converges uniformly to zero, since $\|T_t v\|_\infty \le \|v\|_\infty$. Monotonicity therefore yields $v \le 0$. Removing the mollifier proves comparison at Lebesgue points. Applying it to both signs proves uniqueness. Finally, $G_\lambda \in L^1$ and continuity of its translates in L^1 make $G_\lambda * h$ uniformly continuous. The comparison is therefore pointwise whenever w is continuous. □

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